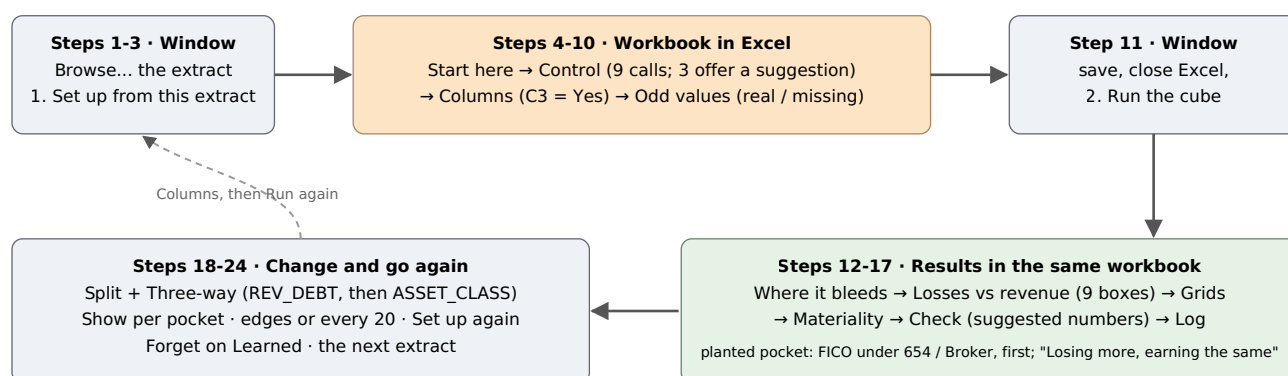


Origination analysis: the window and the workbook

Who this is for: the analyst preparing an origination analysis for a bank's line of business (LOB). No command line is needed. **What you finish with:** one workbook beside the loan extract. It holds your answers, a ranked list of the pockets that lose more than their share, losses set against revenue, and, if you ask for it, every pocket split by a third column and ranked again. **Walked:** 25 September 2026, on commit 5ef61df, with a synthetic book of 8,000 loans. The book hides two things: a bad pocket (scores under 620 in the Broker channel), and borrowers carrying more revolving debt than usual for their score, who go bad 1.8 times as often. RANR has nothing planted in it. No real bank data was used.

What changed since the fourth walk: three Control calls now offer a number worked out from the book (steps 5, 9 and 16). A side of Losses vs revenue counts as more or less only when its own test agrees (step 13). A band width such as every 20 can be typed, and edges are remembered for the next extract (steps 21 and 24). The Three-way tab says on every row what its grid holds fixed (step 18).



Peach = where you answer (Excel). Grey = the window. Green = where you read the results. Nothing is typed at a command line.

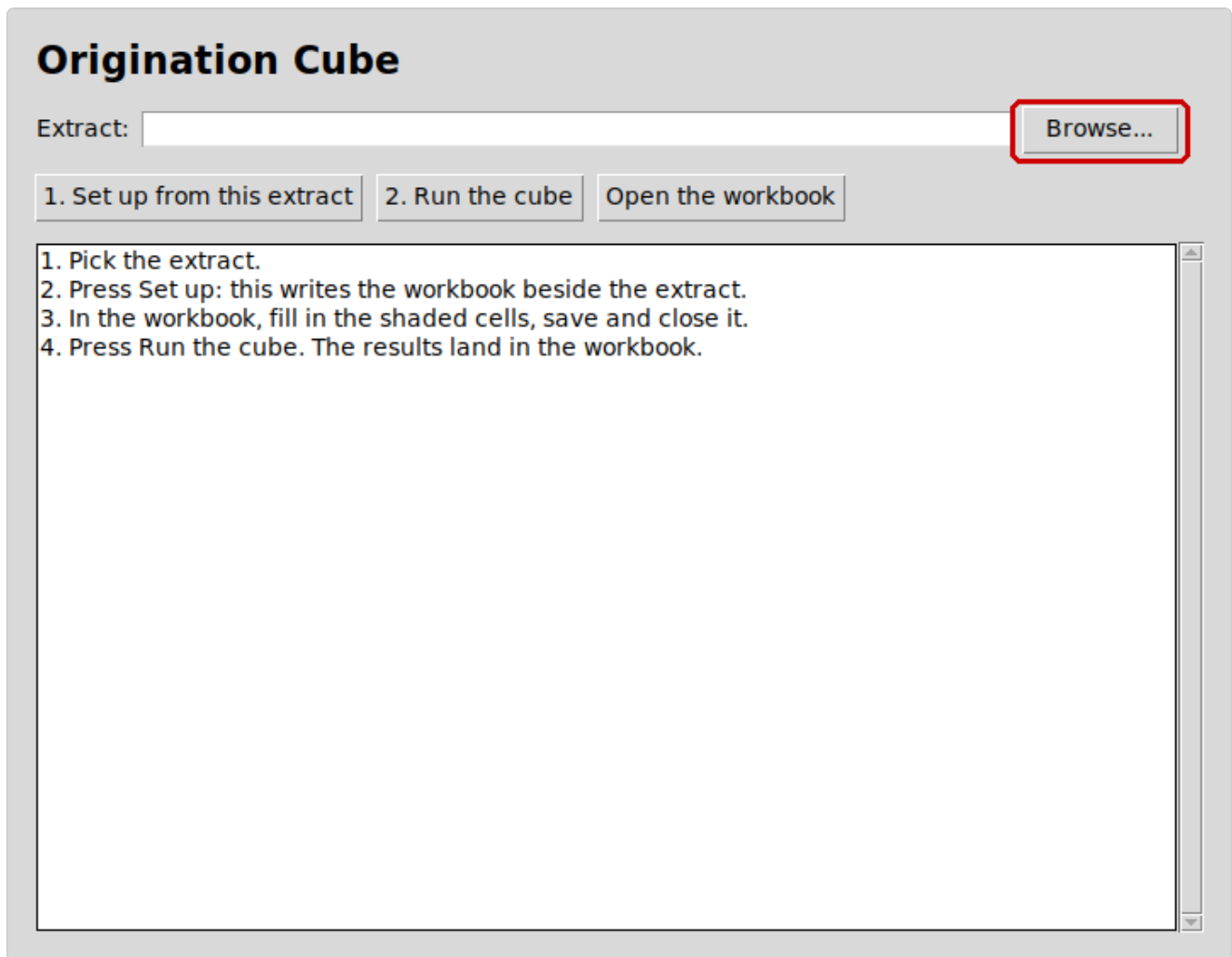
The route: two buttons in the window, one workbook in Excel.

Before you start

- **Python 3.10 or later**, and **Install add-ons.bat** run once. You only need to do this the first time.
- **Excel**.
- **The loan extract** (.csv or .xlsx) in its own folder. It needs a loan number, the booked amount, a yes/no outcome, GCO dollars and RANR dollars.

Step 1: Open the window

Double-click **Origination Cube.pyw** in the origination-cube folder.



Right: a window titled Origination Cube, an Extract box, **Browse...**, three buttons, and four numbered lines in the box underneath.

Step 2: Pick the extract

Press **Browse...** and pick the loan file, not the workbook.

Origination Cube

Extract:

1. Pick the extract.

Directory: /tmp/claude-0/-home-user-SATC/b4305bfc-8769-5da1-bd67-4db589c3cc31/scratchpad/

book testcode
code venv
driver
home
r
shots

File name: /tmp/claude-0/-home-user-SATC/b4305bfc-8769-5da1-bd67-4db589c3cc31/scratchpad/

Files of type: Extracts (*.csv,*.xlsx)

On Windows this is the usual Open dialog. The picture shows the same step on the test machine.

Origination Cube

Extract:

1. Pick the extract.
2. Press Set up: this writes the workbook beside the extract.
3. In the workbook, fill in the shaded cells, save and close it.
4. Press Run the cube. The results land in the workbook.

Right: the file name shows at the right-hand end of the Extract box.

Step 3: Press 1. Set up from this extract

Origination Cube

Extract:

Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv: 8,000 loans, 9 columns.
7 columns to look at first, on the Columns tab.
Next: fill in the shaded cells on Control, Columns and Odd values, save, close, and press Run.

Right: "Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv: 8,000 loans, 9 columns", then "7 columns to look at first", then "Next: fill in the shaded cells on Control, Columns and Odd values".

Check the loan count against what the bank sent. If it's wrong, you picked the wrong file.

Step 4: Open the workbook: Start here

Press **Open the workbook**. It opens on **Start here**.

Origination Cube

Set up from Consumer book Q3.csv: 8,000 loans, 9 columns.

Step	Where	What to do
1	Control	Fill in the shaded cells: materiality, minimum loans, how much worse counts.
2	Columns	Check each column's meaning and whether to cut by it. Set C3 to Yes when done.
3	Odd values	Answer real or missing. Unanswered ones are used as is.
4	Launcher	Save and close this workbook, then press Run the cube.

Where things stand

Calls still to make on Control	9
Things to look at first on Columns	7
Odd values found	2
Last run	not run yet

Right: four steps, and Where things stand: **9** calls to make on Control, 7 things to look at on Columns, 2 odd values, not run yet.

Step 5: Control: see what needs an answer

Click the **Control** tab.

Control center				
Fill in the shaded cells. The rest have starting values; change them if the population calls for it. Pick from the list, or type a number in the next column to override it. Changes apply on the next Run.				
Shaded = still needs an answer				
Setting	Choose	Or enter your own	In use	What it means
What the cube looks at				
How old a loan must be to count				Needs an answer before we run. Pick one, or enter your own.
How columns are recognised				
A number column with this many values or fewer is a category	12 values (recommended)		12 values	Term (36, 48, 60) or a grade from 1 to 10 is read as categories, not cut into bands.
A text column with more values than this is too fine to cut by	50 values (recommended)		50 values	State (50 or so) is still a dimension; a city or a dealer name is raised as a question instead.
How many bands a number column is cut into	5 bands (recommended)		5 bands	Fine enough to see a trend, with each band holding a fifth of the loans.
Where the band edges fall	Each band holds about the same number of loans (recommended)	n/a	Each band holds about the same number of loans	Every band can be tested equally well. Edges land on odd numbers like 663.
Is it enough loans to matter				
Fewest loans in a pocket before it is tested				Needs an answer before we run. Pick one, or enter your own.
Fewest loans with a loss before a loss rate is tested				Needs an answer before we run. Pick one, or enter your own.
Is it material				
Smallest excess loss worth reporting				Needs an answer before we run. Pick one, or enter your own.
Does it underperform				
What a pocket is judged against		n/a		Needs an answer before we run. Pick one from the list.
How much worse than its comparison a pocket must be				Needs an answer before we run. Pick one, or enter your own.
How much better than its comparison a pocket must be				Needs an answer before we run. Pick one, or enter your own.
How far revenue must move before it counts as more or less				Needs an answer before we run. Pick one, or enter your own.
Is it real				
How sure a difference must be before it counts				Needs an answer before we run. Pick one, or enter your own.
How often a real gap should be caught	80% of the time (recommended)		80% of the time	The usual standard. Used only to say how big a gap each pocket's size could show; it never flags anything.
Allowing for testing many pockets at once	Hold down the share of lucky finds (Benjamini-Hochberg) (recommended)	n/a	Hold down the share of lucky finds (Benjamini-Hochberg)	Of the pockets that pass, about 5% at 95% are expected to be luck. Suits scanning for leads.
loans				
Is it enough loans to matter				
Fewest loans in a pocket before it is tested				Needs an answer before we run. Pick one, or enter your own.
Fewest loans with a loss before a loss rate is tested				Needs an answer before we run. Pick one, or enter your own.
Is it material				
Smallest excess loss worth reporting				Needs an answer before we run. Pick one, or enter your own.
Does it underperform				
What a pocket is judged against		n/a		Needs an answer before we run. Pick one from the list.
How much worse than its comparison a				Needs an answer before we run. Pick one, or enter your own.

Right: nine rows are shaded peach. They are the calls the tool won't make for you. Three of them now have an option marked (suggested) in their list, worked out from this book when you Run:

- **Fewest loans in a pocket before it is tested:** Enough for 10 expected losses.
- **How much worse and how much better:** What luck alone can move it.
- **How far revenue must move:** What luck alone can move it.

Nothing is picked for you.

Step 6: Columns: see what each column was taken to be

Click the **Columns** tab.

Columns							
Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.							
Checked every column? ☐							
Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first	Why this was suggested	Blank Samples
LOAN_NBR	Loan number (key)					name contains 'loan'; a different value on every row	0% L0000000, L0000001, L0000002
FICO	FICO score	Yes			160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.	name contains 'fico'; 98% of values between 300 and 850	0% -9999, 713, 716
CHANNEL	Category	Yes			suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.	3 different values	0% Online, Broker, Branch
ORIG_BAL	Booked amount					name contains 'bal'; positive on every loan	0% 7655.75, 27999.47, 39684.43
BAD_FLAG	Outcome (yes/no)	Yes			1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	name contains 'bad'; 0 and 1 (7.1% are 1)	0% 0, 2, 1
GCO_AMT	GCO dollars				1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.	name contains 'gco'; zero on 93% of loans, never negative	0% 0.0, #N/A, 27620.62
RANR_AMT	RANR dollars				2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.	name contains 'ranr'; numbers	0% -165.2, 54.71, 38.01
ASSET_CLASS	Category	Yes			suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.	4 different values	0% 1, 2, 4
REV_DEBT	Amount or number	Yes			suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.	numbers with 7,889 values	0% 15868.86, 12884.7, 8920.97

Columns							
Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.							
Checked every column? ☐							
Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first		
LOAN_NBR	Loan number (key)						
FICO	FICO score	Yes			160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.		
CHANNEL	Category	Yes			suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.		
ORIG_BAL	Booked amount						
BAD_FLAG	Outcome (yes/no)	Yes			1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.		
GCO_AMT	GCO dollars				1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.		
RANR_AMT	RANR dollars				2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.		
ASSET_CLASS	Category	Yes			suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.		
REV_DEBT	Amount or number	Yes			suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.		

Right: one row per column, with a meaning in What it is and the reason beside it. Seven rows are shaded under **Look first**. Checked every column? (C3) is empty. The **Band edges** heading (ringed) now reads 620; 680 or every 20: you can type edges, or a width (step 21).

Step 7: Odd values, and Learned

Click **Odd values**.

Odd values				
Values that might be codes rather than real numbers. Answer real or missing (missing is treated as blank and counted). Unanswered ones are used as is.				
Column	What looks odd	Rows	Answer	Note
FICO	the value -9999 repeated far more than any other	160		
RANR_AMT	negative values in a column that is mostly positive	2,612		

Right: FICO's -9999 (160 rows) and RANR's negatives (2,612 rows), each with an Answer cell.

On the first extract **Learned** says Nothing learned yet.

Learned

Meanings and answers carried over from past runs. Set a row to Forget to drop it on the next Run, and the column waits on Columns for you to confirm it again.

Keep?	Kind	Column	What it learned	First confirmed	Last confirmed	Times
			Nothing learned yet.			

Step 8: If you press Run too early

Save, close and press **2. Run the cube** before answering, and nothing runs.

Origination Cube

Extract: a1-bd67-4db589c3cc31/scratchpad/walk5/book/Consumer book Q3.csv

Browse...

1. Set up from this extract 2. Run the cube Open the workbook

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Control!C6: "How old a loan must be to count" needs an answer. Pick one, or enter your own in column D.
- Control!C13: "Fewest loans in a pocket before it is tested" needs an answer. Pick one, or enter your own in column D.
- Control!C14: "Fewest loans with a loss before a loss rate is tested" needs an answer. Pick one, or enter your own in column D.
- Control!C16: "Smallest excess loss worth reporting" needs an answer. Pick one, or enter your own in column D.
- Control!C18: "What a pocket is judged against" needs an answer. Pick one from the list.
- Control!C19: "How much worse than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
- Control!C20: "How much better than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
- Control!C21: "How far revenue must move before it counts as more or less" needs an answer. Pick one, or enter your own in column D.
- Control!C23: "How sure a difference must be before it counts" needs an answer. Pick one, or enter your own in column D.
- Columns!C3: set Checked every column to Yes once you've checked each column's meaning.

Right: "Couldn't run yet", then one line per thing to fix, naming the tab and cell. All ten fit in the box now. The same list goes to the **Log** tab.

Step 9: Fill in the shaded cells on Control

Click each shaded **Choose** cell and pick from its list, or type a number in **Or enter your own**. The walk took every suggestion:

Setting	Answer
How old a loan must be to count	Every loan

Setting	Answer
Fewest loans in a pocket before it is tested	Enough for 10 expected losses (suggested)
Fewest loans with a loss before a loss rate is tested	10 losses
Smallest excess loss worth reporting	1% of the book's total losses
What a pocket is judged against	The rest of its band
How much worse	What luck alone can move it (suggested)
How much better	What luck alone can move it (suggested)
How far revenue must move before it counts	What luck alone can move it (suggested)
How sure	95% sure

Control center				
Fill in the shaded cells. The rest have starting values; change them if the population calls for it. Pick from the list, or type a number in the next column to override it. Changes apply on the next Run.				
Shaded = still needs an answer				
Setting	Choose	Or enter your own	In use	What it means
What the cube looks at				
How old a loan must be to count	Every loan		Every loan	New loans have not had time to go bad, so a pocket full of recent loans looks better than it is.
How columns are recognised				
A number column with this many values or fewer is a category	12 values (recommended)		12 values	Term (36, 48, 60) or a grade from 1 to 10 is read as categories, not cut into bands.
A text column with more values than this is too fine to cut by	50 values (recommended)		50 values	State (50 or so) is still a dimension; a city or a dealer name is raised as a question instead.
How many bands a number column is cut into	5 bands (recommended)		5 bands	Fine enough to see a trend, with each band holding a fifth of the loans.
Where the band edges fall	Each band holds about the same number of loans (recommended)	n/a	Each band holds about the same number of loans	Every band can be tested equally well. Edges land on odd numbers like 663.
Is it enough loans to matter				
Fewest loans in a pocket before it is tested	Enough for 10 expected losses (suggested)		Enough for 10 expected losses (suggested)	Worked out from this book: the loans a pocket needs to expect 10 with the outcome at the book's rate. Ten is the usual floor for a rate test.
Fewest loans with a loss before a loss rate is tested	10 losses		10 losses	A loss rate rests on how many loans lost money, not how many a pocket holds. Ten is a common floor.
Is it material				
Smallest excess loss worth reporting	1% of the book's total losses		1% of the book's total losses	Scales with the book, so the same setting works on any population.
Does it underperform				
What a pocket is judged against	The rest of its band	n/a	The rest of its band	Broker under 620 against the rest of under 620. Both comparisons are printed; this one decides the flag.
How much worse than its comparison a pocket must be	What luck alone can move it (suggested)		What luck alone can move it (suggested)	Worked out from this book: the gap luck alone can make in a pocket of typical size. A smaller gap can't be told from noise.
How much better than its comparison a pocket must be	What luck alone can move it (suggested)		What luck alone can move it (suggested)	The same as above, the other way: one over the gap luck alone can make in a pocket of typical size.
How far revenue must move before it counts as more or less	What luck alone can move it (suggested)		What luck alone can move it (suggested)	Worked out from this book: the revenue gap luck alone can make in a pocket of typical size. Revenue inside that gap reads as about the same.
Is it real				
How sure a difference must be before it counts	95% sure		95% sure	The usual standard. About 1 pocket in 20 with no real difference will still pass.
How often a real gap should be caught	80% of the time (recommended)		80% of the time	The usual standard. Used only to say how big a gap each pocket's size could show; it never flags anything.
Allowing for testing many pockets at once	Hold down the share of lucky finds (Benjamini-Hochberg) (recommended)	n/a	Hold down the share of lucky finds (Benjamini-Hochberg)	Of the pockets that pass, about 5% at 95% are expected to be luck. Suits scanning for leads.
Is it enough loans to matter				
Fewest loans in a pocket before it is tested	Enough for 10 expected losses (suggested)		Enough for 10 expected losses (suggested)	Worked out from this book: the loans a pocket needs to expect 10 with the outcome at the book's rate. Ten is the usual floor for a rate test.
Fewest loans with a loss before a loss rate is tested	10 losses		10 losses	A loss rate rests on how many loans lost money, not how many a pocket holds. Ten is a common floor.
Is it material				
Smallest excess loss worth reporting	1% of the book's total losses		1% of the book's total losses	Scales with the book, so the same setting works on any population.
Does it underperform				
What a pocket is judged against	The rest of its band	n/a	The rest of its band	Broker under 620 against the rest of under 620. Both comparisons are printed; this one decides the flag.
How much worse than its comparison a pocket must be	What luck alone can move it (suggested)		What luck alone can move it (suggested)	Worked out from this book: the gap luck alone can make in a pocket of typical size. A smaller gap can't be told from noise.
How much better than its comparison a pocket must be	What luck alone can move it (suggested)		What luck alone can move it (suggested)	The same as above, the other way: one over the gap luck alone can make in a pocket of typical size.
How far revenue must move before it counts as more or less	What luck alone can move it (suggested)		What luck alone can move it (suggested)	Worked out from this book: the revenue gap luck alone can make in a pocket of typical size. Revenue inside that gap reads as about the same.
Is it real				

Right: the shading is gone, and In use and What it means fill in.

The suggested numbers don't show here. In use repeats the option's name, "(suggested)" included, before and after a Run. The numbers are on **Check** (step 16).

Step 10: Confirm the columns and answer the odd values

On **Columns**, check each What it is, then set **Checked every column?** (C3) to **Yes**.

Columns

For any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done, confirmed meanings carry over to the next extract.

Checked every column? Yes

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)					name contains 'loan'; a different value on every row	0%	L0000000, L0000001, L0000002
FICO	FICO score	Yes			160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.	name contains 'fico'; 98% of values between 300 and 850	0%	-9999, 713, 716
CHANNEL	Category	Yes			suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.	3 different values	0%	Online, Broker, Branch
ORIG_BAL	Booked amount	Yes			1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	name contains 'bal'; positive on every loan	0%	7655.75, 27999.47, 39684.43
BAD_FLAG	Outcome (yes/no)				1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.	name contains 'bad'; 0 and 1 (7.1% are 1)	0%	0, 2, 1
GCO_AMT	GCO dollars				2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.	name contains 'gco'; zero on 93% of loans, never negative	0%	0.0, #N/A, 27620.62
RANR_AMT	RANR dollars				suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.	name contains 'ranr'; numbers	0%	-165.2, 54.71, 38.01
ASSET_CLASS	Category	Yes			suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.	4 different values	0%	1, 2, 4
REV_DEBT	Amount or number	Yes				numbers with 7,889 values	0%	15868.86, 12884.7, 8920.97

On **Odd values**, the walk said FICO's -9999 is **missing** and RANR's negatives are **real**.

Odd values

Values that might be codes rather than real numbers. Answer real or missing (missing is treated as blank and counted). Unanswered ones are used as is.

Column	What looks odd	Rows	Answer	Note
FICO	the value -9999 repeated far more than any other	160 missing		
RANR_AMT	negative values in a column that is mostly positive	2,612 real		

Step 11: Save, close, and press 2. Run the cube

Origination Cube

Extract: a1-bd67-4db589c3cc31/scratchpad/walk5/book/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

Right:

- "Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree."
- one Worst for line per measure. For the three loss measures it's **FICO under 654 / CHANNEL Broker**, the planted pocket seen through five equal bands.

The window doesn't say what the suggestions came out at. The Worst for RANR line is noise: the test data plants nothing in RANR.

Step 12: Read Where it bleeds

Where it bleeds																
Pockets losing more than their share (NANO, earning loss), largest first. The flag compares each pocket with the rest of its band. Red: worse. Amber: worse, but could be luck. "Luck alone" is how often a gap this big turns up with no real difference behind it.																
Measure	Band column	Band	Segment column	Segment	Loans	Rate	Book rate	Excess	Excess is in	Material	Vs rest of book	Luck alone	Vs rest of band	Luck alone	Flag (vs the rest of its band)	Smallest gap it could show
Outcome, share of loans	FICO	under 654	CHANNEL	Broker	523	24.47%	7.13%	90.7 loans	yes	yes	4.14x under 0.01%	2.38x under 0.01% worse				1.47x or more
Outcome, share of loans	REV DEBT	17,187 and over	CHANNEL	Broker	307	22.29%	7.13%	76.9 loans	yes	yes	3.65x under 0.01%	2.01x under 0.01% worse				1.48x or more
Outcome, share of loans	FICO	under 654	ASSET CLASS	4	392	20.41%	7.13%	52.1 loans	yes	yes	3.17x under 0.01%	1.54x	3.8% worse			1.55x or more
Outcome, share of loans	REV DEBT	17,187 and over	ASSET CLASS	4	424	18.63%	7.13%	48.8 loans	yes	yes	2.87x under 0.01%	1.41x	8.2% worse, but could be luck			1.53x or more
Outcome, share of loans	REV DEBT	17,187 and over	ASSET CLASS	1	411	14.36%	7.13%	29.7 loans	yes	yes	2.13x under 0.01%	0.98x	95.1% in line			1.53x or more
Outcome, share of loans	FICO	under 654	ASSET CLASS	3	350	15.14%	7.13%	28.1 loans	yes	yes	2.24x under 0.01%	1.01x	98.1% in line			1.58x or more
Outcome, share of loans	REV DEBT	17,187 and over	ASSET CLASS	1	420	13.57%	7.13%	27.1 loans	yes	yes	2.01x	0.02%	0.87x	67.5% in line		1.53x or more
Outcome, share of loans	REV DEBT	17,187 and over	ASSET CLASS	3	380	14.21%	7.13%	26.9 loans	yes	yes	2.10x	0.01%	0.96x	93.2% in line		1.56x or more
Outcome, share of loans	ORIG_BAL	15,449 to under 26,324	CHANNEL	Broker	537	11.73%	7.13%	24.7 loans	yes	yes	1.73x	0.58%	1.58x	5.6% worse, but could be luck		1.46x or more
Outcome, share of loans	REV DEBT	17,187 and over	CHANNEL	Branch	525	11.62%	7.13%	23.6 loans	yes	yes	1.71x	0.14%	0.72x	6.3% better, but could be luck		1.47x or more
Outcome, share of loans	REV DEBT	17,187 and over	CHANNEL	Online	568	10.56%	7.13%	19.5 loans	yes	yes	1.54x	0.75%	0.63x	0.22% better		1.45x or more
Outcome, share of loans	ORIG_BAL	37,951 to under 49,152	ASSET CLASS	4	414	11.59%	7.13%	18.5 loans	yes	yes	1.68x	2.4%	1.79x	6.4% worse, but could be luck		1.53x or more
Outcome, share of loans	FICO	under 654	CHANNEL	Online	531	10.36%	7.13%	17.2 loans	yes	yes	1.50x	1.8%	0.59x	0.04% better		1.47x or more
Outcome, share of loans	FICO	654 to under 686	ASSET CLASS	4	410	11.22%	7.13%	16.8 loans	yes	yes	1.62x	1.2%	1.56x	12.5% worse, but could be luck		1.54x or more
Outcome, share of loans	REV DEBT	13,051 to under 17,187	ASSET CLASS	4	393	11.20%	7.13%	16.0 loans	yes	yes	1.62x	1.5%	0.68x	9.0% better, but could be luck		1.55x or more
Outcome, share of loans	REV DEBT	13,051 to under 17,187	CHANNEL	Broker	553	9.95%	7.13%	15.6 loans	yes	yes	1.58x	1.9%	1.51x	15.2% worse, but could be luck		1.53x or more
Outcome, share of loans	FICO	under 654	CHANNEL	Branch	501	10.18%	7.13%	15.3 loans	yes	yes	1.47x	2.9%	0.59x	0.04% better		1.48x or more
Outcome, share of loans	REV DEBT	17,187 and over	ASSET CLASS	2	385	10.41%	7.13%	14.6 loans	yes	yes	1.57x	2.3%	0.69x	8.2% better, but could be luck		1.55x or more
Outcome, share of loans	ORIG_BAL	15,449 to under 26,324	ASSET CLASS	4	413	10.41%	7.13%	13.6 loans	yes	yes	1.50x	8.6%	1.25x	39.8% in line		1.53x or more
Outcome, share of loans	ORIG_BAL	26,324 to under 37,951	CHANNEL	Broker	531	9.23%	7.13%	11.2 loans	yes	yes	1.32x	20.8%	1.30x	25.5% in line		1.47x or more
Outcome, share of loans	ORIG_BAL	26,324 to under 37,951	ASSET CLASS	4	412	9.71%	7.13%	10.6 loans	yes	yes	1.39x	19.6%	1.80x	7.3% worse, but could be luck		1.53x or more
Outcome, share of loans	ORIG_BAL	15,449 to under 26,324	ASSET CLASS	3	427	9.37%	7.13%	9.6 loans	yes	yes	1.34x	24.3%	1.08x	90.8% in line		1.52x or more
Outcome, share of loans	FICO	654 to under 686	CHANNEL	Branch	503	8.95%	7.13%	9.2 loans	yes	yes	1.28x	16.9%	1.13x	82.1% in line		1.48x or more
Outcome, share of loans	ORIG_BAL	15,449 to under 26,324	ASSET CLASS	1	404	9.16%	7.13%	8.2 loans	yes	yes	1.31x	27.9%	1.04x	97.7% in line		1.54x or more
Outcome, share of loans	ORIG_BAL	49,152 and over	CHANNEL	Broker	520	8.46%	7.13%	6.9 loans	yes	yes	1.20x	44.1%	1.90x	5.2% worse, but could be luck		1.47x or more
Outcome, share of loans	ORIG_BAL	37,951 to under 49,152	ASSET CLASS	3	382	8.90%	7.13%	6.8 loans	yes	yes	1.26x	35.9%	1.19x	59.1% in line		1.56x or more
Outcome, share of loans	ORIG_BAL	under 15,449	CHANNEL	Broker	514	8.37%	7.13%	6.4 loans	yes	yes	1.19x	44.8%	1.42x	20.1% worse, but could be luck		1.47x or more
Outcome, share of loans	FICO	654 to under 686	CHANNEL	Broker	531	8.10%	7.13%	5.2 loans	below the line	below the line	1.15x	42.6%	0.97x	87.3% in line		1.47x or more
Outcome, share of loans	ORIG_BAL	under 15,449	ASSET CLASS	4	425	8.24%	7.13%	4.7 loans	below the line	below the line	1.17x	51.6%	1.34x	33.8% worse, but could be luck		1.53x or more
Outcome, share of loans	ORIG_BAL	37,951 to under 49,152	CHANNEL	Branch	526	7.98%	7.13%	4.5 loans	below the line	below the line	1.13x	57.6%	1.03x	88.8% in line		1.47x or more
Outcome, share of loans	FICO	654 to under 686	ASSET CLASS	3	377	7.82%	7.13%	4.1 loans	below the line	below the line	1.18x	51.8%	1.00x	98.1% in line		1.56x or more

Where it bleeds																
Pockets losing more than their share (NANO, earning loss), largest first. The flag compares each pocket with the rest of its band. Red: worse. Amber: worse, but could be luck. "Luck alone" is how often a gap this big turns up with no real difference behind it.																
Measure	Band column	Band	Segment column	Segment	Loans	Rate	Book rate	Excess	Excess is in	Material	Vs rest of book	Luck alone	Vs rest of band	Luck alone	Flag (vs the rest of its band)	Smallest gap it could show
Outcome, share of loans	FICO	under 654	CHANNEL	Broker	523	24.47%	7.13%	90.7 loans	yes	yes	4.14x under 0.01%	2.38x under 0.01% worse				1.47x or more
Outcome, share of loans	REV DEBT	17,187 and over	CHANNEL	Broker	307	22.29%	7.13%	76.9 loans	yes	yes	3.65x under 0.01%	2.01x under 0.01% worse				1.48x or more
Outcome, share of loans	FICO	under 654	ASSET CLASS	4	392	20.41%	7.13%	52.1 loans	yes	yes	3.17x under 0.01%	1.54x	3.8% worse			1.55x or more

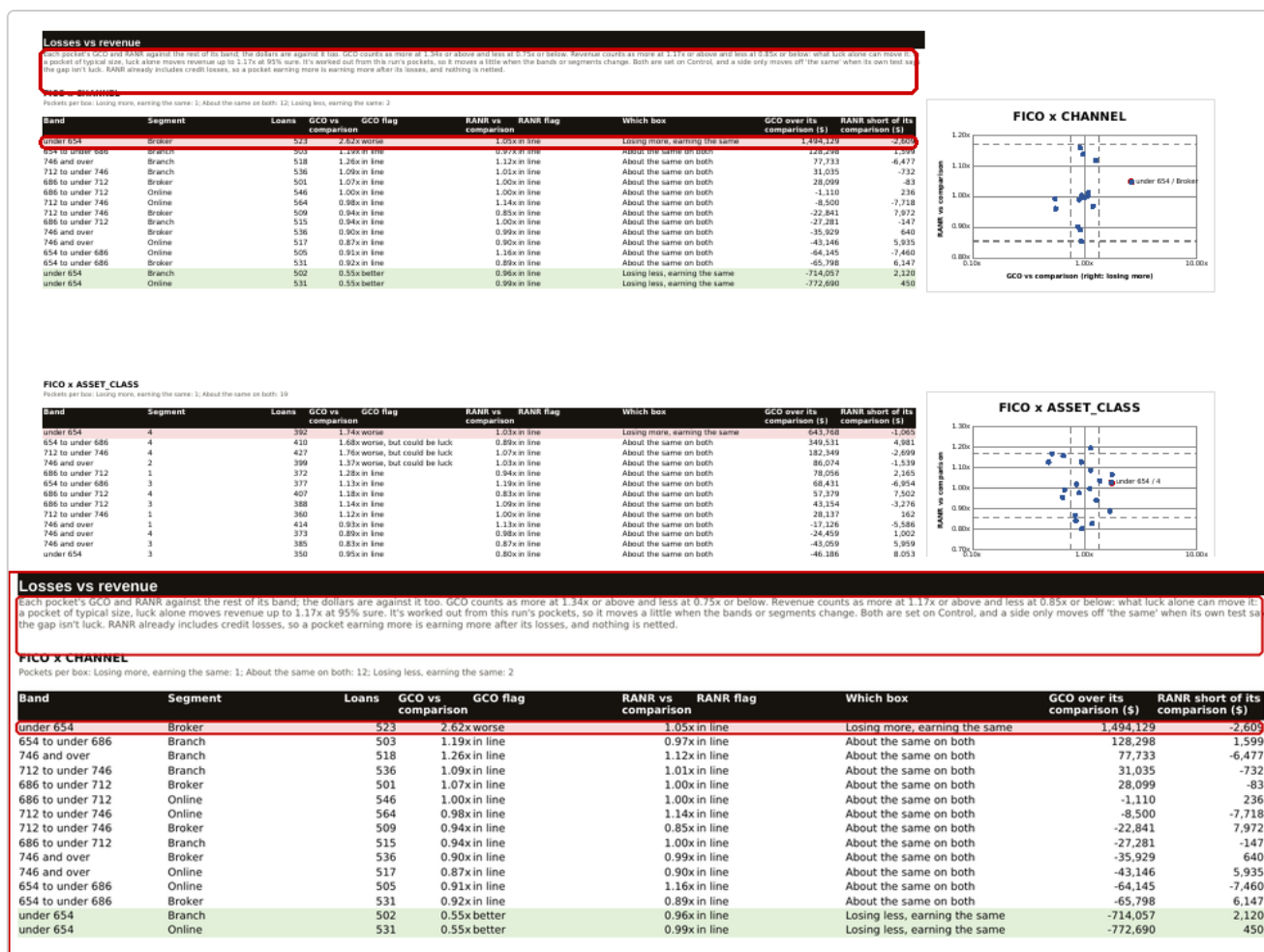
Right: the first row is FICO under 654 / Broker: 523 loans, 24.47% bad against a book of 7.13%, 90.7 loans of excess, 4.14x the rest of the book, 2.38x the rest of its band, **worse**.

How to read a row:

- **Red** rows are worse and the test says it isn't luck. **Amber** rows are worse but could be luck.
- **The flag** uses the comparison chosen in step 9, and the line worked out for how much worse (1.34x here).
- **Luck alone** is how often a gap this big turns up with no real difference.
- **Smallest gap it could show** is what a pocket of that size can reliably find. It's a different figure from the 1.34x line.

Step 13: Losses vs revenue

Every pocket at or above the fewest-loans line (141 here) goes in one of nine boxes. There's one block and one chart per grid.

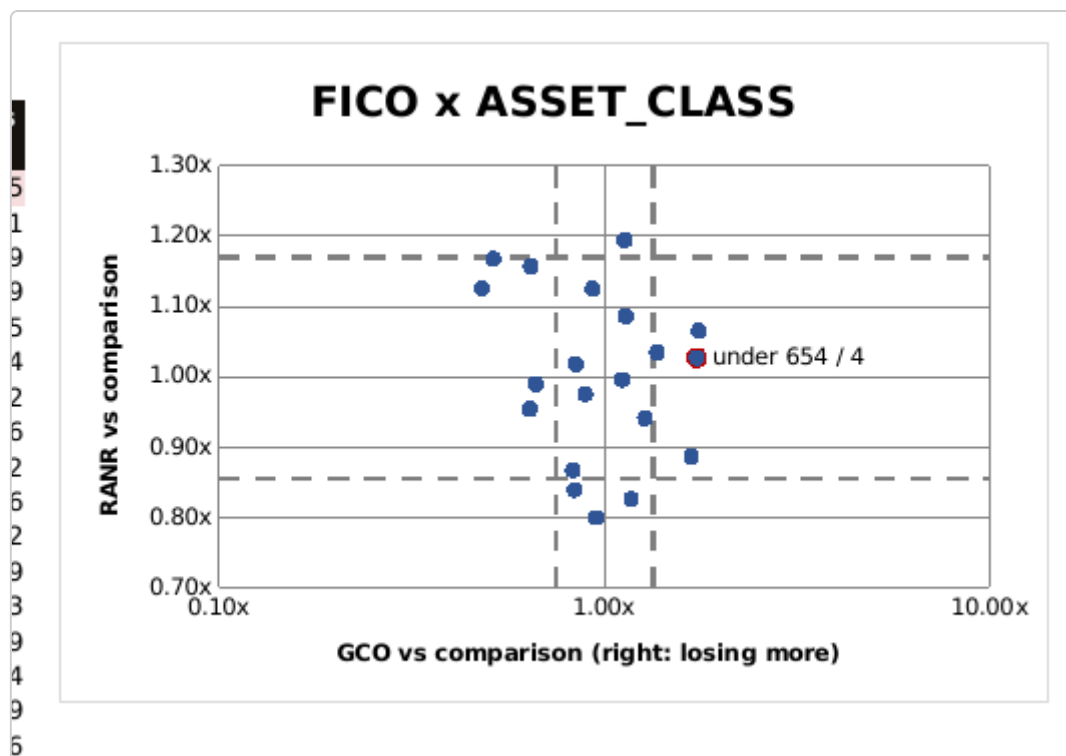


Right:

- The subtitle gives the lines: GCO at **1.34x** and **0.75x**, revenue at **1.17x** and **0.85x** ("what luck alone can move it"). It adds that a side only moves off the same when its own test agrees.
- FICO under 654 / Broker (ringed) reads **Losing more, earning the same**: GCO 2.62x the rest of its band, RANR 1.05x.
- The dollars are against the same comparison as the box, and rows sort by them.
- A pocket with too few losses reads Not tested: too few loans or losses, with no colour.

Read "the same" as "can't tell". A pocket at 1.76x on GCO whose test says could be luck sits in About the same on both (defect 5).

The chart plots the same pockets, with the Control lines dashed. Only pockets in a Losing more box are named:



Dots past a dashed line can still be in "the same". The lines don't decide the box on their own; the test does (defect 3).

Other revenue lines. Change How far revenue must move on Control, save, close and Run. On this book every option gave the same boxes:

Revenue line	Counts as more / less at	Earning more	Earning less
What luck alone can move it	1.17x / 0.85x	0	1
5 percent either way	1.05x / 0.95x	0	1
10 percent either way	1.10x / 0.90x	0	1
The same lines as for losses	1.33x / 0.75x	0	1
Your own: 0.15	1.15x / 0.85x	0	1

The planted pocket reads Losing more, earning the same in every one. With 5% either way, 654 to under 686 / Online at 1.16x still reads earning the same:

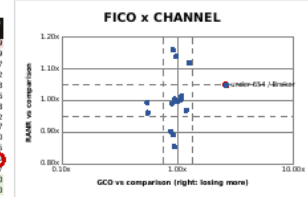
Losses vs revenue

Each pocket's GCO and RANR against the rest of its band: the dollars are against it too. GCO counts as more at 1.34x or above and less at 0.75x or below. Revenue counts as more at 1.05x or above and less at 0.95x or below. 5% either way. Both are set to Control, and a side only moves off 'the same' when its own test says the gap isn't luck. RANR already includes credit losses, so a pocket earning more is earning more after its losses, and nothing is netted.

FICO x CHANNEL

Pockets per box: Losing more, earning the same: 1; About the same on both: 12; Losing less, earning the same: 2

Band	Segment	Loans	GCO vs comparison	GCO flag	RANR vs comparison	RANR flag	Which box	GCO over its comparison (\$)	RANR short of its comparison (\$)
under 654	Broker	523	2.62x worse	1.05x in line	1.05x in line	1.05x in line	Losing more, earning the same	1,494,129	-2,609
654 to under 686	Branch	503	1.19x in line	0.97x in line	0.97x in line	0.97x in line	About the same on both	128,298	1,599
746 and over	Branch	518	1.26x in line	1.12x in line	1.12x in line	1.12x in line	About the same on both	77,733	-6,477
712 to under 746	Branch	536	1.09x in line	1.01x in line	1.01x in line	1.01x in line	About the same on both	31,035	-732
686 to under 712	Broker	501	1.07x in line	1.00x in line	1.00x in line	1.00x in line	About the same on both	28,099	-83
686 to under 712	Online	546	1.00x in line	1.00x in line	1.00x in line	1.00x in line	About the same on both	-1,110	236
712 to under 746	Online	564	0.98x in line	1.14x in line	1.14x in line	1.14x in line	About the same on both	-8,500	-7,718
712 to under 746	Broker	509	0.94x in line	0.85x in line	0.85x in line	0.85x in line	About the same on both	-22,841	7,972
686 to under 712	Branch	515	0.94x in line	1.00x in line	1.00x in line	1.00x in line	About the same on both	-27,281	-147
746 and over	Broker	536	0.90x in line	0.90x in line	0.90x in line	0.90x in line	About the same on both	-43,146	5,935
746 and over	Online	517	0.87x in line	0.90x in line	0.90x in line	0.90x in line	About the same on both	-43,146	5,935
654 to under 686	Online	505	0.91x in line	1.16x in line	1.16x in line	1.16x in line	About the same on both	-64,145	-7,460
Lost to credit side	Online	531	0.92x in line	0.89x in line	0.89x in line	0.89x in line	About the same on both	-65,798	6,147
under 654	Branch	502	0.55x better	0.96x in line	0.96x in line	0.96x in line	Losing less, earning the same	-714,057	2,120
under 654	Online	531	0.55x better	0.99x in line	0.99x in line	0.99x in line	Losing less, earning the same	-772,690	450



Losses vs revenue

Each pocket's GCO and RANR against the rest of its band: the dollars are against it too. GCO counts as more at 1.34x or above and less at 0.75x or below. Revenue counts as more at 1.05x or above and less at 0.95x or below. 5% either way. Both are set to Control, and a side only moves off 'the same' when its own test says the gap isn't luck. RANR already includes credit losses, so a pocket earning more is earning more after its losses, and nothing is netted.

FICO x CHANNEL

Pockets per box: Losing more, earning the same: 1; About the same on both: 12; Losing less, earning the same: 2

Band	Segment	Loans	GCO vs comparison	GCO flag	RANR vs comparison	RANR flag	Which box	GCO over its comparison (\$)	RANR short of its comparison (\$)
under 654	Broker	523	2.62x worse	1.05x in line	1.05x in line	1.05x in line	Losing more, earning the same	1,494,129	-2,609
654 to under 686	Branch	503	1.19x in line	0.97x in line	0.97x in line	0.97x in line	About the same on both	128,298	1,599
746 and over	Branch	518	1.26x in line	1.12x in line	1.12x in line	1.12x in line	About the same on both	77,733	-6,477
712 to under 746	Branch	536	1.09x in line	1.01x in line	1.01x in line	1.01x in line	About the same on both	31,035	-732
686 to under 712	Broker	501	1.07x in line	1.00x in line	1.00x in line	1.00x in line	About the same on both	28,099	-83
686 to under 712	Online	546	1.00x in line	1.00x in line	1.00x in line	1.00x in line	About the same on both	-1,110	236
712 to under 746	Online	564	0.98x in line	1.14x in line	1.14x in line	1.14x in line	About the same on both	-8,500	-7,718
712 to under 746	Broker	509	0.94x in line	0.85x in line	0.85x in line	0.85x in line	About the same on both	-22,841	7,972
686 to under 712	Branch	515	0.94x in line	1.00x in line	1.00x in line	1.00x in line	About the same on both	-27,281	-147
746 and over	Broker	536	0.90x in line	0.90x in line	0.90x in line	0.90x in line	About the same on both	-43,146	5,935
746 and over	Online	517	0.87x in line	0.90x in line	0.90x in line	0.90x in line	About the same on both	-43,146	5,935
654 to under 686	Online	505	0.91x in line	1.16x in line	1.16x in line	1.16x in line	About the same on both	-64,145	-7,460
654 to under 686	Broker	531	0.92x in line	0.89x in line	0.89x in line	0.89x in line	About the same on both	-65,798	6,147
under 654	Branch	502	0.55x better	0.96x in line	0.96x in line	0.96x in line	Losing less, earning the same	-714,057	2,120
under 654	Online	531	0.55x better	0.99x in line	0.99x in line	0.99x in line	Losing less, earning the same	-772,690	450

Your own value is a share up to 0.9. 95 is refused with "for 15%, type 0.15"; 0.95 is refused without a hint.

Origination Cube

Extract: -4db589c3cc31/scratchpad/walk5/wrong/rev-95/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Control!D21: "How far revenue must move before it counts as more or less" needs a share, such as 0.1 for 10%, from 0.01 to 0.9; got 95. Type a share between 0.01 and 0.9: for 15%, type 0.15.

Origination Cube

Extract: 89c3cc31/scratchpad/walk5/wrong/rev-own095/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Control!D21: "How far revenue must move before it counts as more or less" needs a share, such as 0.1 for 10%, from 0.01 to 0.9; got 0.95.

Step 14: Grids

Grids shows each band crossed with each segment, one block per measure. Red is worse, green is better; for RANR low is red.

Grids					Grids					Grids				
FICO x CHANNEL: Outcome, share of loans					Vs the book					Vs the rest of its band				
Rate	Branch	Broker	Online	All	Branch	Broker	Online	All		Branch	Broker	Online	All	
under 654	10.18%	24.47%	10.36%	11.00%	under 654	1.43x	1.43x	1.45x	2.11x	under 654	0.59x	2.38x	2.11x	
654 to under 686	8.95%	8.10%	7.72%	8.25%	654 to under 686	1.26x	1.14x	1.08x	1.16x	654 to under 686	1.13x	0.97x	0.97x	
686 to under 712	5.44%	5.19%	4.21%	4.93%	686 to under 712	0.76x	0.73x	0.59x	0.69x	686 to under 712	1.16x	1.08x	1.08x	
712 to under 746	4.09%	3.14%	3.90%	3.72%	712 to under 746	0.57x	0.44x	0.55x	0.52x	712 to under 746	1.15x	0.79x	0.79x	
746 and over	4.44%	3.73%	3.68%	3.95%	746 and over	0.62x	0.52x	0.52x	0.55x	746 and over	1.20x	0.92x	0.92x	
(blank)					(blank)					(blank)				
(marked missing)	1.96%	12.96%	3.64%	6.25%	(marked missing)	0.91x	1.27x	0.83x	1.00x	(marked missing)				
All	6.47%	9.04%	5.89%	7.13%	All					All				
FICO x CHANNEL: Outcome, share of booked dollars					Vs the book					Vs the rest of its band				
Rate	Branch	Broker	Online	All	Branch	Broker	Online	All		Branch	Broker	Online	All	
under 654	8.43%	21.52%	9.85%	14.92%	under 654	1.43x	1.43x	1.45x	2.11x	under 654	0.59x	2.38x	2.11x	
654 to under 686	8.43%	7.61%	7.42%	7.86%	654 to under 686	1.26x	1.14x	1.08x	1.16x	654 to under 686	1.13x	0.97x	0.97x	
686 to under 712	4.93%	4.63%	4.31%	4.50%	686 to under 712	0.69x	0.68x	0.60x	0.66x	686 to under 712	1.16x	1.08x	1.08x	
712 to under 746	4.12%	3.55%	3.78%	3.62%	712 to under 746	0.64x	0.52x	0.55x	0.52x	712 to under 746	1.15x	0.80x	0.80x	
746 and over	4.48%	3.37%	3.40%	3.68%	746 and over	0.59x	0.49x	0.52x	0.53x	746 and over	1.18x	0.88x	0.97x	
(blank)					(blank)					(blank)				
(marked missing)	1.68%	10.22%	2.93%	4.82%	(marked missing)	0.88x	1.28x	0.82x	1.00x	(marked missing)				
All	6.23%	8.82%	5.69%	6.91%	All					All				
FICO x CHANNEL: GCO per booked dollar					Vs the book					Vs the rest of its band				
Rate	Branch	Broker	Online	All	Branch	Broker	Online	All		Branch	Broker	Online	All	
under 654	5.43%	14.11%	5.43%	8.15%	under 654	1.43x	1.43x	1.45x	2.11x	under 654	0.59x	2.38x	2.11x	
654 to under 686	5.81%	4.23%	4.20%	4.41%	654 to under 686	1.26x	1.14x	1.08x	1.16x	654 to under 686	1.13x	0.97x	0.97x	
686 to under 712	2.42%	2.44%	2.52%	2.53%	686 to under 712	0.69x	0.68x	0.60x	0.66x	686 to under 712	1.16x	1.08x	1.08x	
712 to under 746	2.33%	2.08%	2.14%	2.17%	712 to under 746	0.64x	0.54x	0.56x	0.54x	712 to under 746	1.08x	0.84x	0.86x	
746 and over	2.34%	1.88%	1.77%	1.84%	746 and over	0.59x	0.47x	0.46x	0.50x	746 and over	1.26x	0.86x	0.89x	
(blank)					(blank)					(blank)				
(marked missing)	0.78%	5.93%	1.65%	0.60%	(marked missing)	0.88x	1.38x	0.82x	1.00x	(marked missing)				
All	3.48%	5.52%	3.16%	3.86%	All					All				
FICO x CHANNEL: RANR per booked dollar (more is better)					Vs the book					Vs the rest of its band				
Rate	Branch	Broker	Online	All	Branch	Broker	Online	All		Branch	Broker	Online	All	
under 654	10.18%	24.47%	10.36%	11.00%	under 654	1.43x	1.43x	1.45x	2.11x	under 654	0.59x	2.38x	2.11x	
654 to under 686	8.95%	8.10%	7.72%	8.25%	654 to under 686	1.26x	1.14x	1.08x	1.16x	654 to under 686	1.13x	0.97x	0.97x	
686 to under 712	5.44%	5.19%	4.21%	4.93%	686 to under 712	0.76x	0.73x	0.59x	0.69x	686 to under 712	1.16x	1.08x	1.08x	
712 to under 746	4.09%	3.14%	3.90%	3.72%	712 to under 746	0.57x	0.44x	0.55x	0.52x	712 to under 746	1.15x	0.79x	0.79x	
746 and over	4.44%	3.73%	3.68%	3.95%	746 and over	0.62x	0.52x	0.52x	0.55x	746 and over	1.20x	0.92x	0.92x	
(blank)					(blank)					(blank)				
(marked missing)	1.96%	12.96%	3.64%	6.25%	(marked missing)	0.91x	1.27x	0.83x	1.00x	(marked missing)				
All	6.47%	9.04%	5.89%	7.13%	All					All				

Right: under 654 / Broker is 24.47%, 3.43x the book. The (marked missing) row is blank in the comparison blocks, because it's under the minimums.

Step 15: Materiality

Materiality				
What each materiality level would keep: how many pockets, and how much of the grid's excess they hold. The level itself is set on Control.				
FICO x CHANNEL: Outcome, share of loans				In use: 5.7 loans
Share of the book's loans with the outcome	Excess at or over (loans)	Pockets kept	Share of the grid's excess	
0.5%	2.9	7	100%	
1.0%	5.7	4	92%	
2.0%	11.4	3	86%	
5.0%	28.5	1	63%	
10.0%	57.0	1	63%	
FICO x CHANNEL: Outcome, share of booked dollars				In use: 178,222 ORIG_BAL dollars
Share of the book's ORIG_BAL on loans with the	Excess at or over (ORIG_BAL dollars)	Pockets kept	Share of the grid's excess	
0.5%	89,111	5	97%	
1.0%	178,222	4	94%	
2.0%	356,445	3	88%	
5.0%	891,112	1	65%	
10.0%	1,782,225	1	65%	
FICO x CHANNEL: GCO per booked dollar				In use: 99,554 GCO_AMT dollars
Share of the book's total GCO_AMT	Excess at or over (GCO_AMT dollars)	Pockets kept	Share of the grid's excess	
0.5%	49,777	6	99%	
1.0%	99,554	4	95%	
2.0%	199,107	3	87%	

Right: for FICO x CHANNEL: Outcome, share of loans, the level in use is 5.7 loans (1% of the book's bad loans): 4 pockets, 92% of the excess.

Step 16: Check

Check records what the run did. **This is where the suggested numbers are shown** (ringed).

Check

Settings used, tie-outs, and what was left out.

Extract	Consumer book Q3.csv
Loans run	8,000
Record of this run	Consumer book Q3 - Origination Cube - what ran.yaml, beside this workbook: every setting the run used, kept for the file. It is replaced by the next Run.
Tie-out checks	132 of 132 agree: every grid adds up to the book
Left out of Outcome, share of loans	1 x BAD_FLAG flag not 0 or 1
Left out of Outcome, share of booked dollars	1 x ORIG_BAL blank; 1 x BAD_FLAG flag not 0 or 1
Left out of GCO per booked dollar	1 x GCO_AMT not a number; 1 x ORIG_BAL blank
Left out of RANR per booked dollar	1 x ORIG_BAL blank
Band edges used: FICO	654; 686; 712; 746 (5 bands)
Band edges used: ORIG_BAL	15,449; 26,324; 37,951; 49,152 (5 bands)
Band edges used: REV_DEBT	6,559; 9,837; 13,051; 17,187 (5 bands)
Loans needed for a 1.34x gap: Outcome, share of loans	about 971
Loans needed for a 1.34x gap: Outcome, share of booked dollars	about 1,219
Loans needed for a 1.34x gap: GCO per booked dollar	about 1,305
Loans needed for a 1.34x gap: RANR per booked dollar	about 228
Materiality line: Outcome, share of loans	5.7 loans
Materiality line: Outcome, share of booked dollars	178,222 ORIG_BAL dollars
Materiality line: GCO per booked dollar	99,554 GCO_AMT dollars
Materiality line: RANR per booked dollar	8,288 RANR_AMT dollars
Worked out from this book	fewest loans 141 (enough to expect 10 with the outcome at the book's rate of 7.13%); worse at 1.34x; better at 0.75x (the outcome gap luck alone can make in a pocket of typical size)
The allowance for many tests covers	each grid and measure on its own, one comparison at a time (the firm's call, 25 Sep 2026)
Revenue counts as more or less at	1.17x and 0.85x: what luck alone can move it: in a pocket of typical size, luck alone moves revenue up to 1.17x at 95% sure. It's worked out from this run's pockets, so it moves a little when the bands or segments change
How old a loan must be to count	Every loan
Fewest loans in a pocket before it is tested	141
Fewest loans with a loss before a loss rate is tested	10 losses
Smallest excess loss worth reporting	1% of the book's total losses
What a pocket is judged against	The rest of its band
How much worse than its comparison a pocket must be	1.34
How much better than its comparison a pocket must be	0.75
How far revenue must move before it counts as more or less	What luck alone can move it
How close a difference must be before it counts	95% sure
Materiality line: RANR per booked dollar	8,288 RANR_AMT dollars
Worked out from this book	fewest loans 141 (enough to expect 10 with the outcome at the book's rate of 7.13%); worse at 1.34x; better at 0.75x (the outcome gap luck alone can make in a pocket of typical size)
The allowance for many tests covers	each grid and measure on its own, one comparison at a time (the firm's call, 25 Sep 2026)
Revenue counts as more or less at	1.17x and 0.85x: what luck alone can move it: in a pocket of typical size, luck alone moves revenue up to 1.17x at 95% sure. It's worked out from this run's pockets, so it moves a little when the bands or segments change
How old a loan must be to count	Every loan

Right:

- "Worked out from this book: fewest loans 141 (enough to expect 10 with the outcome at the book's rate of 7.13%); worse at 1.34x; better at 0.75x".
- "Revenue counts as more or less at 1.17x and 0.85x: what luck alone can move it".
- Further down, the settings show 141, 1.34 and 0.75, and the revenue line as What luck alone can move it.
- "132 of 132 agree".

The record file beside the workbook (... - what ran.yaml) has 141, 1.34 and 0.75 as plain numbers. It has the revenue line as luck, with no number.

Step 17: Log, and Start here

Log

Every Run and every refusal, newest first.

2026-09-25 18:58 Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

2026-09-25 18:58 Couldn't run. Fix these, save, close, and press Run again:
Control!C6: "How old a loan must be to count" needs an answer. Pick one, or enter your own in column D.
Control!C13: "Fewest loans in a pocket before it is tested" needs an answer. Pick one, or enter your own in column D.
Control!C14: "Fewest loans with a loss before a loss rate is tested" needs an answer. Pick one, or enter your own in column D.

Control!C16: "Smallest excess loss worth reporting" needs an answer. Pick one, or enter your own in column D.
Control!C18: "What a pocket is judged against" needs an answer. Pick one from the list.
Control!C19: "How much worse than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
Control!C20: "How much better than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.
Control!C21: "How far revenue must move before it counts as more or less" needs an answer. Pick one, or enter your own in column D.
Control!C23: "How sure a difference must be before it counts" needs an answer. Pick one, or enter your own in column D.

Columns!C3: set Checked every column to Yes once you've checked each column's meaning.

Start here now shows 0 calls, 0 things to look at on Columns, and the last run:

Origination Cube

Last run on 8,000 loans from Consumer book Q3.csv.

Step	Where	What to do
1	Control	Fill in the shaded cells: materiality, minimum loans, how much worse counts.
2	Columns	Check each column's meaning and whether to cut by it. Set C3 to Yes when done.
3	Odd values	Answer real or missing. Unanswered ones are used as is.
4	Launcher	Save and close this workbook, then press Run the cube.
Where things stand		
	Calls still to make on Control	0
	Things to look at first on Columns	0
	Odd values found	2
	Last run	2026-09-25 18:58: Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

Step 18: Split every pocket by a number (revolving debt)

On **Columns**, set **Split pockets by it?** to **Yes** on REV_DEBT. Save, close, and Run.

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column? Yes

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)					name contains 'loan'; a different value on every row	0%	L0000000, L0000001, L0000002
FICO	FICO score	Yes			160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.	name contains 'fico'; 98% of values between 300 and 850	0%	-9999, 713, 716
CHANNEL	Category	Yes			suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.	3 different values	0%	Online, Broker, Branch
ORIG_BAL BAD_FLAG	Booked amount Outcome (yes/no)	Yes			1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	name contains 'bal'; positive on every loan name contains 'bad'; 0 and 1 (7.1% are 1)	0%	7655.75, 27999.47, 39684.43 0, 2, 1
GCO_AMT	GCO dollars				1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.	name contains 'gco'; zero on 93% of loans, never negative	0%	0.0, #N/A, 27620.62
RANR_AMT	RANR dollars				2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.	name contains 'ranr'; numbers	0%	-165.2, 54.71, 38.01
ASSET_CLASS	Category	Yes			suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.	4 different values	0%	1, 2, 4
REV_DEBT	Amount or number	Yes		Yes	suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.	numbers with 7,889 values	0%	15868.86, 12884.7, 8920.97

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column? Yes

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first
LOAN_NBR	Loan number (key)				
FICO	FICO score	Yes			160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.
CHANNEL	Category	Yes			suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.
ORIG_BAL BAD_FLAG	Booked amount Outcome (yes/no)	Yes			1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.
GCO_AMT	GCO dollars				1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.
RANR_AMT	RANR dollars				2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.
ASSET_CLASS	Category	Yes			suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.
REV_DEBT	Amount or number	Yes		Yes	suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.

Origination Cube

Extract:

Ran on 8,000 loans from Consumer book Q3.csv; 581 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Nothing is worse for RANR per booked dollar at these settings.
Split by REV_DEBT: each pocket halved at its own median. See the Split and Three-way tabs.
REV_DEBT isn't cut on its own while it splits.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

Right: "581 tie-out checks agree" and "Split by REV_DEBT: each pocket halved at its own median."

On **Split**, each paragraph starts with what its grid holds fixed:

Split

Every pocket split in two at its own median REV_DEBT: the high half against the low half, holding the band and segment fixed. Above 1.00x means the high half does worse (for RANR, earns more). Grids that hold fixed what REV_DEBT moves with come first. REV_DEBT isn't cut into bands of its own while it splits.

FICO x CHANNEL: Outcome, share of loans

This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed. REV_DEBT's correlation with ORIG_BAL is -0.02: 0 means unrelated, and the further from 0, the more of this gap may be ORIG_BAL. The high-REV_DEBT half was worse in 14 of the 15 pockets big enough to test. Holding the pocket fixed, the high half has the outcome 1.95 times as often as the low half (95% range 1.72 to 2.19). Luck alone gives a gap this big under 0.01% of the time. As odds, 2.10x (1.75 to 2.53). The gap is about the same size in every pocket.

High half vs low	Branch	Broker	Online
under 654	2.39x	2.67x	2.63x
654 to under 686	1.37x	2.92x	2.26x
686 to under 712	2.12x	1.90x	1.56x
712 to under 746	2.14x	0.78x	1.44x
746 and over	1.30x	2.33x	1.12x
(blank)			
(marked missing)			

Luck alone	Branch	Broker	Online
under 654	0.18% under 0.01%	6.2%	
654 to under 686	26.9%	0.07%	1.2%
686 to under 712	5.0%	10.5%	28.7%
712 to under 746	8.1%	61.8%	38.5%
746 and over	52.3%	6.8%	80.9%
(blank)			
(marked missing)			

FICO x CHANNEL: Outcome, share of booked dollars

This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed. REV_DEBT's correlation with ORIG_BAL is -0.02: 0 means unrelated, and the further from 0, the more of this gap may be ORIG_BAL. The high-REV_DEBT half was worse in 15 of the 15 pockets big enough to test. Holding the pocket fixed, the high half loses 2.07x what it would at the low half's rate (95% range 1.80 to 2.34). Luck alone gives a gap this big under 0.01% of the time.

High half vs low	Branch	Broker	Online
under 654	2.84x	2.11x	2.05x
654 to under 686	1.58x	3.82x	2.83x
686 to under 712	2.39x	2.69x	1.30x
712 to under 746	2.68x	1.12x	3.63x
746 and over	1.17x	2.82x	1.01x
(blank)			
(marked missing)			

Luck alone	Branch	Broker	Online
under 654	0.09% under 0.01%	1.7%	
654 to under 686	15.9%	0.04%	0.44%
686 to under 712	5.1%	3.2%	56.4%
712 to under 746	4.6%	84.1%	30.2%
746 and over	73.2%	5.2%	98.8%
(blank)			
(marked missing)			

FICO x CHANNEL: GCO per booked dollar

This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed. REV_DEBT's correlation with ORIG_BAL is -0.02: 0 means unrelated, and the further from 0, the more of this gap may be ORIG_BAL. The high-REV_DEBT half was worse in 14 of the 15 pockets big enough to test. Holding the pocket fixed, the high half loses 2.06x what it would at the low half's rate (95% range 1.78 to 2.33). Luck alone gives a gap this big under 0.01% of the time.

Split

Every pocket split in two at its own median REV_DEBT: the high half against the low half, holding the band and segment fixed. Above 1.00x means the high half does worse (for RANR, earns more). Grids that hold fixed what REV_DEBT moves with come first. REV_DEBT isn't cut into bands of its own while it splits.

FICO x CHANNEL: Outcome, share of loans

This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed. REV_DEBT's correlation with ORIG_BAL is -0.02: 0 means unrelated, and the further from 0, the more of this gap may be ORIG_BAL. The high-REV_DEBT half was worse in 14 of the 15 pockets big enough to test. Holding the pocket fixed, the high half has the outcome 1.95 times as often as the low half (95% range 1.72 to 2.19). Luck alone gives a gap this big under 0.01% of the time. As odds, 2.10x (1.75 to 2.53). The gap is about the same size in every pocket.

High half vs low	Branch	Broker	Online
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Luck alone	Branch	Broker	Online
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- For FICO x CHANNEL: "This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53)", then the high-debt half at **1.95 times** the low half (95% range 1.72 to 2.19). The planted effect is 1.8x.
- The ORIG_BAL grids open "This grid holds ORIG_BAL fixed (... -0.02). It doesn't hold FICO fixed", then say 2.55x. **Don't quote them.** On a book with no debt effect they still say 1.62x, "under 0.01%" (defect 4).

ORIG_BAL x CHANNEL: Outcome, share of loans

This grid holds ORIG_BAL fixed (REV_DEBT's correlation with it: -0.02). It doesn't hold FICO fixed. REV_DEBT's correlation with FICO is -0.53: 0 means unrelated, and the further from 0, the more of this gap may be FICO. The high-REV_DEBT half was worse in 15 of the 15 pockets big enough to test. Holding the pocket fixed, the high half has the outcome 2.55 times as often as the low half (95% range 2.26 to 2.83). Luck alone gives a gap this big under 0.01% of the time. As odds, 2.73x (2.26 to 3.29). The gap is about the same size in every pocket.

High half vs low	Branch	Broker	Online
under 15,449	1.92x	4.38x	1.90x
15,449 to under 26,324	1.47x	2.72x	1.63x
26,324 to under 37,951	1.76x	5.85x	5.00x
37,951 to under 49,152	3.67x	2.51x	2.41x

Luck alone	Branch	Broker	Online
under 15,449	5.4% under 0.01%	8.6%	
15,449 to under 26,324	22.8% under 0.01%	10.8%	
26,324 to under 37,951	10.3% under 0.01%	0.01%	
37,951 to under 49,152	under 0.01%	0.15%	1.2%

Three-way ranks every band / segment / half pocket, FICO grids first within each measure:

Three-way

Pockets split by REV_DEBT, each tested like any other pocket, grids that hold fixed what it moves with first, being more than their share (RANR, earning less), larger first. The top compares each pocket with the rest of its band. Red: worse. Another worse, but could be luck. "Luck alone" is how often a gap this big turns up with no real difference behind it.

Measure	Band column Band	Segment column Segment	Loans	Rate	Book rate	Excess	Excess to in	Material	Va rest of book	Luck alone	Va rest of band	Luck alone	Flag (vs the rest of its band)	Smallest gap it could show	What this grid holds fixed
Outcome, share of loans	FICO	under 654	CHANNEL / REV_DEBT	BRKER / REV_DEBT high half	261	25.93%	7.13%	74.0 loans	yes	0.78% under 0.01%	2.71% under 0.01% under	2.38x or more ORIG_BAL			This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed. REV_DEBT's correlation with ORIG_BAL is -0.02: 0 means unrelated, and the further from 0, the more of this gap may be ORIG_BAL.
Outcome, share of loans	FICO	under 654	ASSET_CLASS / REV_DEBT	4 / REV_DEBT high half	196	27.55%	7.13%	40.0 loans	yes	4.17% under 0.01%	2.88%	0.04% worse		1.80x or more ORIG_BAL	This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed. REV_DEBT's correlation with ORIG_BAL is -0.02: 0 means unrelated, and the further from 0, the more of this gap may be ORIG_BAL.
Outcome, share of loans	FICO	under 654	ASSET_CLASS / REV_DEBT	1 / REV_DEBT high half	210	20.48%	7.13%	28.0 loans	yes	3.03% under 0.01%	1.44%	14.2% worse, but could be luck		1.77x or more ORIG_BAL	This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed. REV_DEBT's correlation with ORIG_BAL is -0.02: 0 means unrelated, and the further from 0, the more of this gap may be ORIG_BAL.
Outcome, share of loans	FICO	under 654	ASSET_CLASS / REV_DEBT	3 / REV_DEBT high half	175	20.80%	7.13%	22.5 loans	yes	2.82% under 0.01%	1.39%	23.7% worse, but could be luck		1.85x or more ORIG_BAL	This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed. REV_DEBT's correlation with ORIG_BAL is -0.02: 0 means unrelated, and the further from 0, the more of this gap may be ORIG_BAL.

The last column, **What this grid holds fixed**, carries the same sentence on every row:

is big turns up with no real difference behind it.

Flag (vs the rest of its band)	Smallest gap it could show	What this grid holds fixed
1% worse	1.68x or more	This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed. REV_DEBT's correlation with ORIG_BAL is -0.02: 0 means unrelated, and the further from 0, the more of this gap may be ORIG_BAL.
4% worse	1.80x or more	This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed. REV_DEBT's correlation with ORIG_BAL is -0.02: 0 means unrelated, and the further from 0, the more of this gap may be ORIG_BAL.
2% worse, but could be luck	1.77x or more	This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed. REV_DEBT's correlation with ORIG_BAL is -0.02: 0 means unrelated, and the further from 0, the more of this gap may be ORIG_BAL.
7% worse, but could be luck	1.85x or more	This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed. REV_DEBT's correlation with ORIG_BAL is -0.02: 0 means unrelated, and the further from 0, the more of this gap may be ORIG_BAL.
0% is less	1.70x or more	This grid holds FICO fixed (REV_DEBT's correlation with it: -0.53). It doesn't hold ORIG_BAL fixed. REV_DEBT's correlation with ORIG_BAL is -0.02: 0 means unrelated, and the further from 0, the more of this gap may be ORIG_BAL.

Step 19: Split every pocket by a category (asset class)

Clear REV_DEBT's split and set **ASSET_CLASS** instead. Save, close, and Run.

Origination Cube

Extract: a1-bd67-4db589c3cc31/scratchpad/walk5/book/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv; 382 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Worst for RANR per booked dollar: REV DEBT 6,559 to under 9,837 / CHANNEL Broker.
Split by ASSET_CLASS: one layer per value. See the Three-way tab. ASSET_CLASS isn't cut on its own while it splits.
Open Consumer book Q3 - Origination Cube.xlsx. start with Where it bleeds.

Right: "382 tie-out checks agree" and "Split by ASSET_CLASS: one layer per value."

Three-way																
Pockets split by ASSET_CLASS, each tested like any other pocket, losing more than their share (RANK, earning less, target first. The flag compares each pocket with the rest of its band. Red: worse, Amber: worse, but could be luck. "Luck alone" is how often a gap this big turns up with no real difference behind it.																
Measure	Band column Band		Segment column	Segment	Loans	Rate	Book rate	Excess is in	Material	Vs rest of book	Luck alone	Vs rest of band	Luck alone	Flag (vs the rest of its band)	Smallest gap it could show	
Outcome, share of loans	FICO	under 654	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 4	131	29.01%	7.13%	28.7 loans	yes	4.29x	2.11x	too few loans to test	4.99x or more			
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 4	124	27.42%	7.13%	25.2 loans	yes	4.03x	2.02x	too few loans to test	2.02x or more			
Outcome, share of loans	FICO	under 654	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 3	111	29.73%	7.13%	25.1 loans	yes	4.37x	2.14x	too few loans to test	2.09x or more			
Outcome, share of loans	FICO	under 654	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 1	145	22.07%	7.13%	21.7 loans	yes	3.22x under 0.01%	1.54x	16.1% worse, but could be luck	1.94x or more			
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 1	131	23.66%	7.13%	21.7 loans	yes	3.45x	1.71x	too few loans to test	1.99x or more			
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 3	120	24.17%	7.13%	20.4 loans	yes	3.52x	1.74x	too few loans to test	2.04x or more			
Outcome, share of loans	FICO	under 654	CHANNEL / ASSET_CLASS	Broker / ASSET_CLASS 2	136	18.38%	7.13%	15.3 loans	yes	2.65x	1.25x	too few loans to test	1.97x or more			
Outcome, share of loans	REV_DEBT	17,187 and over	CHANNEL / ASSET_CLASS	Branch / ASSET_CLASS 4	146	15.75%	7.13%	12.6 loans	yes	2.26x	1.09x	99.4% in line	1.94x or more			
Outcome, share of loans	FICO	under 654	CHANNEL / ASSET_CLASS	Branch / ASSET_CLASS 4	130	16.15%	7.13%	11.7 loans	yes	2.32x	1.08x	too few loans to test	2.00x or more			
Outcome, share of loans	FICO	under 654	CHANNEL / ASSET_CLASS	Online / ASSET_CLASS 4	131	16.03%	7.13%	11.7 loans	yes	2.30x	1.07x	too few loans to test	1.99x or more			

With the suggested fewest loans (141), most three-way pockets aren't tested.

FICO under 654 / Broker / asset class 4 has 131 loans and 29% bad, and reads too few loans to test. 60 of the 77 bad-loan rows read the same, and none is flagged worse. With 30 loans, the fourth walk had this pocket flagged worse (defect 1).

Step 20: Show a median or average per pocket

On **Columns**, set **Show per pocket** to average on ORIG_BAL and median on REV_DEBT. Save, close, and Run.

FICO x CHANNEL: average ORIG_BAL per pocket

Beside the rates, for reading them: not tested, and an average doesn't add up across pockets.

Average	Branch	Broker	Online	All
under 654	32,549	32,583	32,707	32,614
654 to under 686	32,159	32,621	31,389	32,066
686 to under 712	32,699	32,967	32,345	32,661
712 to under 746	30,751	31,887	32,569	31,747
746 and over	31,757	31,970	32,765	32,162
(blank)		27,999		27,999
(marked missing)	34,726	30,898	33,613	33,052
All	32,022	32,370	32,390	32,263

FICO x CHANNEL: median REV_DEBT per pocket

Beside the rates, for reading them: not tested, and a median doesn't add up across pockets.

Median	Branch	Broker	Online	All
under 654	17,209	16,277	17,083	16,904
654 to under 686	13,556	13,291	13,669	13,535
686 to under 712	12,292	12,169	12,233	12,219
712 to under 746	9,952	10,431	10,659	10,272
746 and over	7,430	6,911	7,234	7,205
(blank)		12,885		12,885
(marked missing)	9,964	10,431	10,267	10,326
All	11,443	11,394	11,328	11,393

FICO x ASSET_CLASS: Outcome, share of loans

Loans where BAD_FLAG is 1, as a share of all loans.

Rate	1	2	3	4	All
under 654	13.57%	11.20%	15.14%	20.41%	15.05%
654 to under 686	7.30%	5.92%	8.22%	11.22%	8.25%
686 to under 712	4.57%	3.04%	5.41%	6.63%	4.93%
712 to under 746	3.05%	3.48%	2.55%	5.62%	3.72%
746 and over	3.62%	4.01%	4.16%	4.02%	3.95%
(blank)		0.00%			0.00%

Right: "Beside the rates, for reading them: not tested, and an average doesn't add up across pockets."

Step 21: Your own band edges, or a band width

Edges: type 620; 680; 740 in FICO's **Band edges** cell, with semicolons. Save, close, and Run.

Origination Cube

Extract: a1-bd67-4db589c3cc31/scratchpad/walk5/book/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

Worst for Outcome, share of loans: FICO under 620 / CHANNEL Broker.

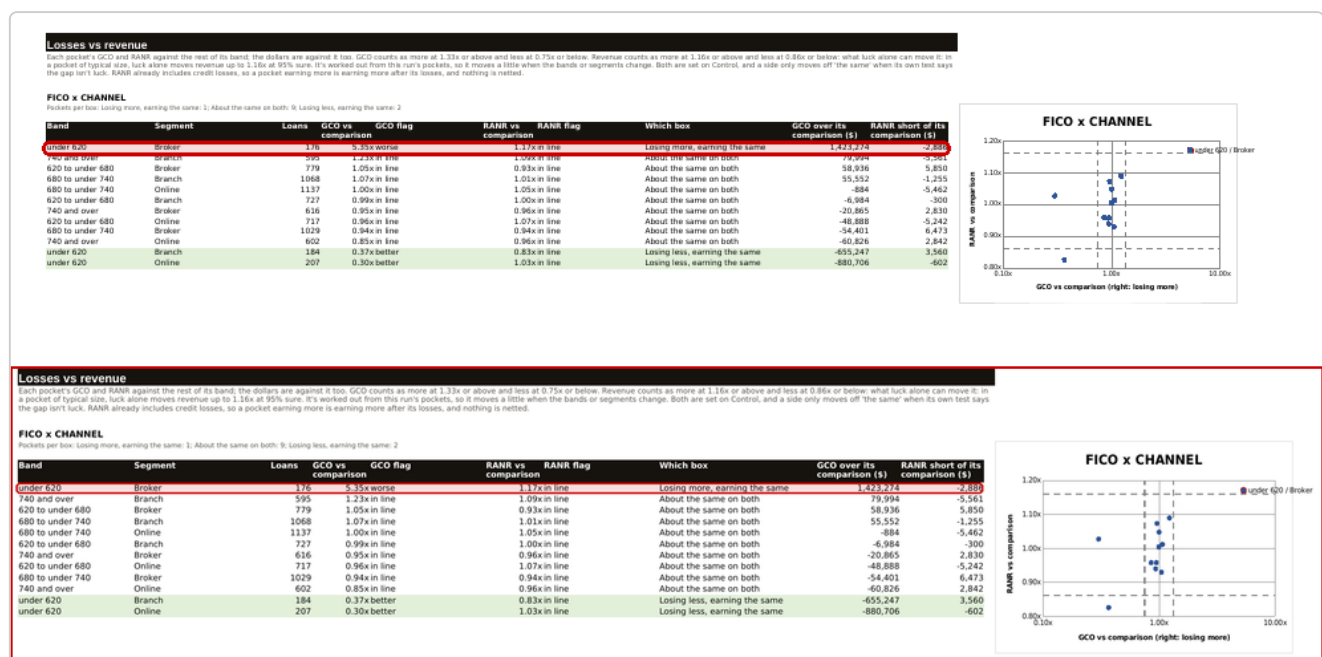
Worst for Outcome, share of booked dollars: FICO under 620 / CHANNEL Broker.

Worst for GCO per booked dollar: FICO under 620 / CHANNEL Broker.

Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.

Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

Right: the worst pocket is **FICO under 620 / Broker**. On Losses vs revenue it reads **Losing more, earning the same**: 176 loans, GCO 5.35x the rest of its band, RANR 1.17x. The revenue line on this run is 1.16x, and the RANR test says in line, so the side stays at the same:



A width: type every 20 instead.

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column? Yes

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)							
FICO	FICO score	Yes	every 20		160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.	name contains 'loan'; a different value on every row	0%	L0000000, L0000001, L0000002
CHANNEL	Category	Yes			suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.	name contains 'fico'; 98% of values between 300 and 850	0%	-9999, 713, 716
ORIG_BAL BAD_FLAG	Booked amount Outcome (yes/no)	Yes		average	1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	3 different values	0%	Online, Broker, Branch
GCO_AMT	GCO dollars				1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.	name contains 'bal'; positive on every loan	0%	7655.75, 27999.47, 39684.43
RANR_AMT	RANR dollars				2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.	name contains 'bad'; 0 and 1 (7.1% are 1)	0%	0, 2, 1
ASSET_CLASS	Category	Yes			suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.	name contains 'gco'; zero on 93% of loans, never negative	0%	0.0, #N/A, 27620.62
REV_DEBT	Amount or number	Yes		median	suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.	name contains 'ranr'; numbers	0%	-165.2, 54.71, 38.01
						4 different values	0%	1, 2, 4
						numbers with 7,889 values	0%	15868.86, 12884.7, 8920.97

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column? Yes

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first
LOAN_NBR	Loan number (key)				
FICO	FICO score	Yes	every 20		160 rows look odd (repeated value). Used as it is until you answer it on the Odd values tab.
CHANNEL	Category	Yes			suggested as Category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.
ORIG_BAL BAD_FLAG	Booked amount Outcome (yes/no)	Yes		average	1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.
GCO_AMT	GCO dollars				1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.
RANR_AMT	RANR dollars				2,612 rows look odd (negatives). Used as it is until you answer it on the Odd values tab.
ASSET_CLASS	Category	Yes			suggested as Category from its shape alone (4 different values). If it's something more specific, say so and it will be remembered.
REV_DEBT	Amount or number	Yes		median	suggested as Amount or number from its shape alone (numbers with 7,889 values). If it's something more specific, say so and it will be remembered.

Origination Cube

Extract: a1-bd67-4db589c3cc31/scratchpad/walk5/book/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

Worst for Outcome, share of loans: REV_DEBT 17,187 and over / CHANNEL Broker.

Worst for Outcome, share of booked dollars: REV_DEBT 17,187 and over / CHANNEL Broker.

Worst for GCO per booked dollar: REV_DEBT 17,187 and over / CHANNEL Broker.

Nothing is worse for RANK per booked dollar at these settings.

Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

Check

Settings used, tie-outs, and what was left out.

Extract	Consumer book Q3.csv
Loans run	8,000
Record of this run	Consumer book Q3 - Origination Cube - what ran.yaml, beside this workbook: every setting the run used, kept for the file. It is replaced by the next Run.
Tie-out checks	132 of 132 agree: every grid adds up to the book
Left out of Outcome, share of loans	1 x BAD_FLAG flag not 0 or 1
Left out of Outcome, share of booked dollars	1 x ORIG_BAL blank; 1 x BAD_FLAG flag not 0 or 1
Left out of GCO per booked dollar	1 x GCO_AMT not a number; 1 x ORIG_BAL blank
Left out of RANK per booked dollar	1 x ORIG_BAL blank
Left out of average ORIG_BAL	1 x ORIG_BAL blank
Band edges used: FICO	500; 520; 540; 560; 580; 600; 620; 640; 660; 680; 700; 720; 740; 760; 780; 800; 820; 840; 860; 880; 900; 920 (23 bands)
Band edges used: ORIG_BAL	15,449; 26,324; 37,951; 49,152 (5 bands)
Band edges used: REV_DEBT	6,559; 9,837; 13,051; 17,187 (5 bands)

Left out of average ORIG_BAL	1 x ORIG_BAL blank
Band edges used: FICO	500; 520; 540; 560; 580; 600; 620; 640; 660; 680; 700; 720; 740; 760; 780; 800; 820; 840; 860; 880; 900; 920 (23 bands)
Band edges used: ORIG_BAL	15,449; 26,324; 37,951; 49,152 (5 bands)

Right: Check lists 22 edges, 500 to 920 (23 bands).

What the window says: the worst pocket is now REV_DEBT 17,187 and over / Broker, and FICO isn't named. With the suggested fewest loans, the planted pocket's pieces (600 to 620 / Broker: 99 loans, 50.5% bad) read too few loans to test (defect 1). If you use 20-point bands, set **Fewest loans** yourself.

Refused widths: every 1 and every 0.5 ("would make 61 bands"), every 0, every -5 and every twenty ("should read like every 20"). See wrong turns P to R.

Step 22: Press Set up again

Press **1. Set up from this extract** again.

Origination Cube

Extract: Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv: 8,000 loans, 9 columns.
2 columns to look at first, on the Columns tab.
Everything is answered. Press Run the cube.

Right: "Everything is answered. Press Run the cube." Control, Columns and Odd values answers are kept, and so are the last results.

Origination Cube		
Set up from Consumer book Q3.csv: 8,000 loans, 9 columns.		
Step	Where	What to do
1	Control	Fill in the shaded cells: materiality, minimum loans, how much worse counts.
2	Columns	Check each column's meaning and whether to cut by it. Set C3 to Yes when done.
3	Odd values	Answer real or missing. Unanswered ones are used as is.
4	Launcher	Save and close this workbook, then press Run the cube.
Where things stand		
	Calls still to make on Control	0
	Things to look at first on Columns	2
	Odd values found	2
	Last run	2026-09-25 19:05: Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

Look at the Band edges column before you Run. Edges remembered from other workbooks are written into empty cells. On this walk REV_DEBT picked up every 2000 and CHANNEL 620; 680 from wrong turns run on other copies. Nothing on the tab or in the window says so (defect 2):

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column? Yes

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)							
FICO	FICO score	Yes	every 20			Remembered: you confirmed this as loan or application number on 2026-09-25	0%	L0000000, L0000001, L0000002
CHANNEL	Category	Yes	620; 680			Remembered: you confirmed this as a FICO score on 2026-09-25	0%	-9999, 713, 716
ORIG_BAL	Booked amount	Yes		average		Remembered: you confirmed this as a category to cut by: channel, state, product on 2026-09-25	0%	Online, Broker, Branch
BAD_FLAG	Outcome (yes/no)					Remembered: you confirmed this as booked or loan amount, which weights the reporting-level rates on 2026-09-25	0%	7655.75, 27999.47, 39684.43
GCO_AMT	GCO dollars					1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	0%	0, 2, 1
RANR_AMT	RANR dollars					1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.	0%	0.0, #N/A, 27620.62
ASSET_CLASS	Category	Yes				Remembered: you confirmed this as gross charge-off dollars on 2026-09-25	0%	-165.2, 54.71, 38.01
REV_DEBT	Amount or number	Yes	every 2000	median		Remembered: you confirmed this as RANR dollars on 2026-09-25	0%	1, 2, 4
						Remembered: you confirmed this as a number to cut into bands on 2026-09-25	0%	15868.86, 12884.7, 8920.97

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column? Yes

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first
LOAN_NBR	Loan number (key)				
FICO	FICO score	Yes	every 20		
CHANNEL	Category	Yes	620; 680		
ORIG_BAL	Booked amount	Yes		average	
BAD_FLAG	Outcome (yes/no)				
GCO_AMT	GCO dollars				
RANR_AMT	RANR dollars				
ASSET_CLASS	Category	Yes			
REV_DEBT	Amount or number	Yes	every 2000	median	

Step 23: Prune something learned

On **Learned**, set a row's **Keep?** (column A) to **Forget**. Save, close and Run.

Learned

Meanings and answers carried over from past runs. Set a row to Forget to drop it on the next Run, and the column waits on Columns for you to confirm it again.

Keep?	Kind	Column	What it learned	First confirmed	Last confirmed	Times
Keep	column	ASSET_CLASS	Category	2026-09-25	2026-09-25	23
Keep	column	BAD_FLAG	Outcome (yes/no)	2026-09-25	2026-09-25	23
Forget	column	CHANNEL	category; band edges 620; 680	2026-09-25	2026-09-25	23
Keep	column	FICO	fico; band edges every 20	2026-09-25	2026-09-25	23
Keep	column	GCO_AMT	GCO dollars	2026-09-25	2026-09-25	23
Keep	column	LOAN_NBR	Loan number (key)	2026-09-25	2026-09-25	23
Keep	column	ORIG_BAL	Booked amount	2026-09-25	2026-09-25	23
Keep	column	RANR_AMT	RANR dollars	2026-09-25	2026-09-25	23
Keep	column	REV_DEBT	amount; band edges every 2000	2026-09-25	2026-09-25	23

Right: the window says "Forgot CHANNEL, as marked on Learned. Check it on Columns and set C3 to Yes before the next Run."

Origination Cube

Extract:

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
Nothing is worse for Outcome, share of loans at these settings.
Worst for Outcome, share of booked dollars: ORIG_BAL 49,152 and over / CHANNEL Broker.
Worst for GCO per booked dollar: ORIG_BAL 49,152 and over / CHANNEL Broker.
Nothing is worse for RANR per booked dollar at these settings.
Forgot CHANNEL, as marked on Learned. Check it on Columns and set C3 to Yes before the next Run.
~~Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.~~

The next Run refuses until you do:

Origination Cube

Extract:

[Browse...](#)

1. Set up from this extract

2. Run the cube

[Open the workbook](#)

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!C3: set Checked every column to Yes once you've checked each column's meaning. Forgotten on Learned: CHANNEL. Check what it is, then set C3 to Yes again.

On **Columns**, C3 is empty with "Forgotten on Learned: CHANNEL. Check what it is, then set C3 to Yes again."

Columns								
Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.								
Checked every column? Forgotten on Learned: CHANNEL. Check what it is, then set C3 to Yes again.								
Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)					Remembered: you confirmed this as loan or application number on 2026-09-25	0%	L0000000, L0000001, L0000002
FICO	FICO score	Yes	every 20			Remembered: you confirmed this as a FICO score on 2026-09-25	0%	-9999, 713, 716
CHANNEL	Category	Yes	620; 680		Forgotten on Learned: confirm what it is.	Forgotten on Learned: it was remembered before.	0%	Online, Broker, Branch
ORIG_BAL	Booked amount	Yes		average		Remembered: you confirmed this as booked or loan amount, which weights the reporting-level rates on 2026-09-25	0%	7655.75, 27999.47, 39684.43
BAD_FLAG	Outcome (yes/no)				1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	Remembered: you confirmed this as the yes/no outcome: charged off, ever delinquent, decided by the system on 2026-09-25	0%	0, 2, 1
GCO_AMT	GCO dollars				1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.	Remembered: you confirmed this as gross charge-off dollars on 2026-09-25	0%	0.0, #N/A, 27620.62
RANR_AMT	RANR dollars					Remembered: you confirmed this as RANR dollars on 2026-09-25	0%	-165.2, 54.71, 38.01
ASSET_CLASS	Category	Yes				Remembered: you confirmed this as a category to cut by: channel, state, product on 2026-09-25	0%	1, 2, 4
REV_DEBT	Amount or number	Yes	every 2000	median		Remembered: you confirmed this as a number to cut into bands on 2026-09-25	0%	15868.86, 12884.7, 8920.97

Columns								
Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.								
Checked every column? Forgotten on Learned: CHANNEL. Check what it is, then set C3 to Yes again.								
Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)					Remembered: you confirmed this as loan or application number on 2026-09-25	0%	L0000000, L0000001, L0000002
FICO	FICO score	Yes	every 20			Remembered: you confirmed this as a FICO score on 2026-09-25	0%	-9999, 713, 716
CHANNEL	Category	Yes	620; 680		Forgotten on Learned: confirm what it is.	Forgotten on Learned: it was remembered before.	0%	Online, Broker, Branch

Forget is the only way to drop remembered band edges. Clearing the cell doesn't: the next Set up fills it in again (defect 2). Forget drops the column's meaning too.

Step 24: The next extract

Browse to the next quarter's extract and press **Set up**.

Origination Cube

Extract:

Set up Consumer book Q4 - Origination Cube.xlsx from Consumer book Q4.csv: 6,000 loans, 9 columns.
2 columns to look at first, on the Columns tab.
Next: fill in the shaded cells on Control and Columns, save, close, and press Run.

Right: "2 columns to look at first". Meanings are remembered, and so are the band edges typed last time. FICO comes in as every 20:

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column?

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)							
FICO	FICO score	Yes	every 20			Remembered: you confirmed this as loan or application number on 2026-09-25	0%	L0000000, L0000001, L0000002
CHANNEL	Category	Yes				Remembered: you confirmed this as a FICO score on 2026-09-25	0%	-9999, 694, 713
ORIG_BAL	Booked amount	Yes				Remembered: you confirmed this as a category to cut by: channel, state, product on 2026-09-25	0%	Broker, Branch, Online
BAD_FLAG	Outcome (yes/no)					Remembered: you confirmed this as booked or loan amount, which weights the reporting-level rates on 2026-09-25	0%	29850.78, 29562.62, 58061.68
GCO_AMT	GCO dollars					Remembered: you confirmed this as the yes/no outcome: charged off, ever delinquent, decided by the system on 2026-09-25	0%	0, 2, 1
RANR_AMT	RANR dollars					Remembered: you confirmed this as gross charge-off dollars on 2026-09-25	0%	0.0, #N/A, 6604.12
ASSET_CLASS	Category	Yes				Remembered: you confirmed this as RANR dollars on 2026-09-25	0%	107.15, 285.79, 299.02
REV_DEBT	Amount or number	Yes				Remembered: you confirmed this as a category to cut by: channel, state, product on 2026-09-25	0%	2, 4, 1
						Remembered: you confirmed this as a number to cut into bands on 2026-09-25	0%	18812.23, 6699.74, 10834.7

Columns

Fix any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column?

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; Show per 680 or every 20) pocket	Split pockets by it?	Look first
LOAN_NBR	Loan number (key)				
FICO	FICO score	Yes	every 20		
CHANNEL	Category	Yes			
ORIG_BAL	Booked amount	Yes			
BAD_FLAG	Outcome (yes/no)				1 value is neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.
GCO_AMT	GCO dollars				1 value isn't a number among 5,999 that are. They're left out of any rate using this column and counted.
RANR_AMT	RANR dollars				
ASSET_CLASS	Category	Yes			
REV_DEBT	Amount or number	Yes			

Control starts blank again. The judgment calls are asked every time.

When something goes wrong

What you did	What the window says	What to do
A. Split set on two columns	"Columns!H13 and Columns!H14: only one column can split the pockets. Clear all but one."	Clear one.
B. Split the only category that's cut	"Nothing is left to cut across ... CHANNEL (Columns!H8: it splits the pockets ...); ASSET_CLASS (Columns!D13, Cut by it? No)."	Clear the split, or set ASSET_CLASS back to Yes.
C. average on a category	"Columns!G8: "CHANNEL" isn't a number column, so it has no average to show."	Clear the cell.
D. Edges 620,680,740 turned into one number by Excel	"... read as the single number 620,680,740. Excel dropped the commas. Type them with semicolons"	Retype with semicolons.
D2. An edge outside the column (620; 680; 950)	"FICO runs from 496 to 921, so an edge at 950 would leave a band empty."	Type edges inside that range.
E. Workbook still open in Excel	"... is open in Excel. Close it, then press Run again." Set up says the same. Memory unchanged.	Close Excel and press again.

What you did	What the window says	What to do
F. A column renamed after Set up	"Couldn't run: the extract has no column "CHANNEL" (a segment) ... press Set up again." Set up then says "New columns since the last check: CHNL."	Press Set up, check the new name, set C3 to Yes.
G. Fewer rows after Set up	"Consumer book Q3.csv has changed since Set up ...", then it runs on 6,000.	Check the loan count.
H. Workbook copied to a folder with a different extract of the same name	The same "has changed" line; it runs the new folder's 3,000 loans.	Press Set up in the new folder first.
I. Split set on GCO, the key or the outcome	"Columns!H11: "GCO_AMT" is marked GCO dollars, which can't split the pockets. Only a score, ratio, amount (the booked amount too) or category can."	Split by something known at booking.
I2. Split by the booked amount	Runs, 571 tie-outs.	As the message says, it's allowed.
J. A dollar amount typed as materiality (100000)	Runs.	Check says it's GCO dollars only.
K. The workbook picked as the extract	Set up: "... is the workbook, not the loan file. Pick the extract it was set up from". Run: "Pick the extract beside it."	Browse to the .csv.
L. 95 typed for "how sure"	"... from 0.5 to 0.999; got 95. For 95%, type 0.95."	Type 0.95.
M. 95, or 0.95, as your own revenue line	95: "... for 15%, type 0.15." 0.95: refused, no hint.	Type a share up to 0.9.
N. New columns in a refreshed extract	"New columns since the last check: CURR_STATUS, DTI. Columns!C3 needs a Yes again."	Check them, set C3 to Yes.
O. Forget on Learned, then Run twice	First Run runs and says what was forgotten; the second refuses on C3.	Check the column, set C3 to Yes.
P. every 1 or every 0.5 on FICO	"Columns: every 1 on "FICO" would make 61 bands (498 to 889). Use a wider band, 50 at most."	Use a wider width. The 61 is wrong (defect 7).
Q. every 0, every -5, every twenty	"Columns!F7: "every 0" should read like every 20: the word every, then how wide each band is."	Type a positive number in figures.
R. Edges or a width typed on a category (CHANNEL)	Runs; the cell is ignored, and remembered.	Clear it, and Forget the column on Learned.
S. Your own fewest loans bigger than any pocket (3000), with luck suggested	Runs, "Nothing is worse". Check says "Worked out from this book: worse at 1.25x" (defect 6).	Use a smaller number.

A: two splits

Origination Cube

Extract:

[Browse...](#)

[1. Set up from this extract](#)

[2. Run the cube](#)

[Open the workbook](#)

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:
- Columns!H13 and Columns!H14: only one column can split the pockets. Clear all but one.

B: the only segment split

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Nothing is left to cut across: mark at least one column as a category (or term) on the Columns tab, with Cut by it set to Yes. Changed from what was suggested: CHANNEL (Columns!H8: it splits the pockets, so it isn't a segment of its own); ASSET_CLASS (Columns!D13, Cut by it? No).

C: average on a category

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!G8: "CHANNEL" isn't a number column, so it has no average to show. Clear the cell, or change what it is.

D and D2: edges

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!F7: the band edges for "FICO" read as the single number 620,680,740. Excel dropped the commas. Type them with semicolons: 620; 680; 740.

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:
- Columns!F7: FICO runs from 496 to 921, so an edge at 950 would leave a band empty. Type edges inside that range, with semicolons: 620; 680; 740.

E: workbook open. Simulated by locking the file (chattr +i), since Excel isn't available here.

Origination Cube

Extract:

Consumer book Q3 - Origination Cube.xlsx is open in Excel. Close it, then press Run again.

Origination Cube

Extract:

Consumer book Q3 - Origination Cube.xlsx is open in Excel. Close it, then press Set up again.

F: a column renamed, then Set up

Origination Cube

Extract:

[Browse...](#)

[1. Set up from this extract](#)

[2. Run the cube](#)

[Open the workbook](#)

Couldn't run: the extract has no column "CHANNEL" (a segment). Its columns are: LOAN_NBR, FICO, CHNL, ORIG_BAL, BAD_FLAG, GCO_AMT, RANR_AMT, ASSET_CLASS, REV_DEBT. If a column was renamed or dropped, press Set up again.

Origination Cube

Extract:

[Browse...](#)

1. Set up from this extract

2. Run the cube

[Open the workbook](#)

Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv: 8,000 loans, 9 columns.

New columns since the last check: CHNL. Columns!C3 needs a Yes again.

3 columns to look at first, on the Columns tab.

Next: fill in the shaded cells on Columns, save, close, and press Run.

G: fewer rows

Origination Cube

Extract:

Consumer book Q3.csv has changed since Set up. If columns were added or renamed, press Set up first.

Ran on 6,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.

Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.

Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.

Worst for RANR per booked dollar: FICO 714 to under 747 / ASSET_CLASS 2.

Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

H: the workbook copied to another folder

Origination Cube

Extract:

Consumer book Q3.csv has changed since Set up. If columns were added or renamed, press Set up first.

Ran on 3,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.

Worst for Outcome, share of loans: ORIG_BAL 48,955 and over / ASSET_CLASS 4.

Worst for Outcome, share of booked dollars: ORIG_BAL 48,955 and over / ASSET_CLASS 4.

Worst for GCO per booked dollar: ORIG_BAL 48,955 and over / ASSET_CLASS 4.

Nothing is worse for RANR per booked dollar at these settings.

Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

I and I2: splits

Origination Cube

Extract:

[Browse...](#)

1. Set up from this extract

2. Run the cube

[Open the workbook](#)

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!H11: "GCO_AMT" is marked GCO dollars, which can't split the pockets. Only a score, ratio, amount (the booked amount too) or category can.

Origination Cube

Extract:

[Browse...](#)

[1. Set up from this extract](#)

[2. Run the cube](#)

[Open the workbook](#)

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!H6: "LOAN_NBR" is marked Loan number (key), which can't split the pockets. Only a score, ratio, amount (the booked amount too) or category can.

Origination Cube

Extract:

[Browse...](#)

[1. Set up from this extract](#)

[2. Run the cube](#)

[Open the workbook](#)

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!H10: "BAD_FLAG" is marked Outcome (yes/no), which can't split the pockets. Only a score, ratio, amount (the booked amount too) or category can.

Origination Cube

Extract:

[Browse...](#)

1. Set up from this extract

2. Run the cube

[Open the workbook](#)

Ran on 8,000 loans from Consumer book Q3.csv; 571 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
Split by ORIG_BAL: each pocket halved at its own median. See the Split and Three-way tabs.
ORIG_BAL isn't cut on its own while it splits.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

J: your own dollar materiality

Origination Cube

Extract:

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
Worst for Outcome, share of loans: FICO under 654 / CHANNEL Broker.
Worst for Outcome, share of booked dollars: FICO under 654 / CHANNEL Broker.
Worst for GCO per booked dollar: FICO under 654 / CHANNEL Broker.
Worst for RANR per booked dollar: REV_DEBT 6,559 to under 9,837 / CHANNEL Broker.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

K: the workbook picked, then Run

Origination Cube

Extract:

[Browse...](#)

[1. Set up from this extract](#)

[2. Run the cube](#)

[Open the workbook](#)

Consumer book Q3 - Origination Cube.xlsx is the workbook, not the loan file. Pick the extract it was set up from (Consumer book Q3.csv or Consumer book Q3.xlsx).

Origination Cube

Extract:

[Browse...](#)

[1. Set up from this extract](#)

[2. Run the cube](#)

[Open the workbook](#)

Consumer book Q3 - Origination Cube.xlsx is the workbook, not the loan file. Pick the extract beside it.

L: 95 for "how sure"

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:
- Control!D23: "How sure a difference must be before it counts" needs a share, such as 0.95 for 95%, from 0.5 to 0.999; got 95. For 95%, type 0.95.

N: new columns

Origination Cube

Extract: 9c3cc31/scratchpad/walk5/wrong/new-columns/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv: 8,000 loans, 11 columns.

New columns since the last check: CURR_STATUS, DTI. Columns!C3 needs a Yes again.

4 columns to look at first, on the Columns tab.

Next: fill in the shaded cells on Columns, save, close, and press Run.

Columns

For any meaning that's wrong, and set Cut by it to No for anything you don't want in the grids. Shaded rows have a reason under Look first. Set C3 to Yes when done; confirmed meanings carry over to the next extract.

Checked every column? **New since the last check: CURR_STATUS, DTI. Check them, then set C3 to Yes again.**

Column	What it is	Cut by it? Yes means (outcome only)	Band edges (620; 680 or every 20)	Show per pocket	Split pockets by it?	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	Loan number (key)								
FICO	FICO score	Yes	every 20				Remembered: you confirmed this as loan or application number on 2026-09-25	0%	L0000000, L0000001, L0000002
CHANNEL	Category	Yes					Remembered: you confirmed this as a FICO score on 2026-09-25	0%	-9999, 713, 716
ORIG_BAL	Booked amount	Yes					Remembered: you confirmed this as a category to cut by: channel, state, product on 2026-09-25	0%	Online, Broker, Branch
BAD_FLAG	Outcome (yes/no)						Remembered: you confirmed this as booked or loan amount, which weights the reporting-level rates on 2026-09-25	0%	7655.75, 27999.47, 39684.43
GCO_AMT	GCO dollars						Remembered: you confirmed this as the yes/no outcome: charged off, ever delinquent, decided by the system on 2026-09-25	0%	0, 2, 1
RANR_AMT	RANR dollars						1 value is neither 0 nor 1. They're left out of the outcome rates and counted, if they mean something, fix the extract.	0%	0.0, #N/A, 27620.62
ASSET_CLASS	Category	Yes					1 value isn't a number among 7,999 that are. They're left out of any rate using this column and counted.	0%	-165.2, 54.71, 38.01
REV_DEBT	Amount or number	Yes	every 2000				Remembered: you confirmed this as RANR dollars on 2026-09-25	0%	-165.2, 54.71, 38.01
CURR_STATUS	Servicing data						Remembered: you confirmed this as a category to cut by: channel, state, product on 2026-09-25	0%	1, 2, 4
DTI	DTI-style ratio	Yes					Remembered: you confirmed this as a number to cut into bands on 2026-09-25	0%	15868.86, 12884.7, 8920.97
							New since the last check.	0%	Current, CO, 300PD
							name contains 'status'	0%	
							New since the last check.	0%	0.346, 0.235, 0.287
							name contains 'dti'; a ratio with median 30%	0%	

P: every 1, every 0.5

Origination Cube

Extract:

[Browse...](#)

[1. Set up from this extract](#)

[2. Run the cube](#)

[Open the workbook](#)

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns: every 1 on "FICO" would make 61 bands (498 to 889). Use a wider band, 50 at most.

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:
- Columns: every 0.5 on "FICO" would make 61 bands (498 to 889). Use a wider band, 50 at most.

Q: every 0, every twenty

Origination Cube

Extract:

[Browse...](#)

1. Set up from this extract

2. Run the cube

[Open the workbook](#)

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!F7: "every 0" should read like every 20: the word every, then how wide each band is.

Origination Cube

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!F7: "every twenty" should read like every 20: the word every, then how wide each band is.

R: a width on a category

Origination Cube

Extract:

Ran on 6,000 loans from Consumer book Q4.csv; 132 tie-out checks agree.
Worst for Outcome, share of loans: REV_DEBT 17,344 and over / CHANNEL Broker.
Worst for Outcome, share of booked dollars: REV_DEBT 17,344 and over / CHANNEL Broker.
Worst for GCO per booked dollar: REV_DEBT 17,344 and over / CHANNEL Broker.
Nothing is worse for RANR per booked dollar at these settings.
Open Consumer book Q4 - Origination Cube.xlsx: start with Where it bleeds.

S: a fewest-loans number bigger than any pocket

Origination Cube

Extract:

[Browse...](#)

[1. Set up from this extract](#)

[2. Run the cube](#)

[Open the workbook](#)

Ran on 8,000 loans from Consumer book Q3.csv; 132 tie-out checks agree.
Nothing is worse for Outcome, share of loans at these settings.
Nothing is worse for Outcome, share of booked dollars at these settings.
Nothing is worse for GCO per booked dollar at these settings.
Nothing is worse for RANR per booked dollar at these settings.
Open Consumer book Q3 - Origination Cube.xlsx: start with Where it bleeds.

Check

Settings used, tie-outs, and what was left out.

Extract	Consumer book Q3.csv
Loans run	8,000
Record of this run	Consumer book Q3 - Origination Cube - what ran.yaml, beside this workbook: every setting the run used, kept for the file. It is replaced by the next Run.
Tie-out checks	132 of 132 agree: every grid adds up to the book
Left out of Outcome, share of loans	1 x BAD_FLAG flag not 0 or 1
Left out of Outcome, share of booked dollars	1 x ORIG_BAL blank; 1 x BAD_FLAG flag not 0 or 1
Left out of GCO per booked dollar	1 x GCO_AMT not a number; 1 x ORIG_BAL blank
Left out of RANR per booked dollar	1 x ORIG_BAL blank
Band edges used: FICO	654; 686; 712; 746 (5 bands)
Band edges used: ORIG_BAL	15,449; 26,324; 37,951; 49,152 (5 bands)
Band edges used: REV_DEBT	6,559; 9,837; 13,051; 17,187 (5 bands)
Loans needed for a 1.25x gap: Outcome, share of loans	about 1,756
Loans needed for a 1.25x gap: Outcome, share of booked dollars	about 2,203
Loans needed for a 1.25x gap: GCO per booked dollar	about 2,359
Loans needed for a 1.25x gap: RANR per booked dollar	about 411
Materiality line: Outcome, share of loans	5.7 loans
Materiality line: Outcome, share of booked dollars	178,222 ORIG_BAL dollars
Materiality line: GCO per booked dollar	99,554 GCO_AMT dollars
Materiality line: RANR per booked dollar	8,288 RANR_AMT dollars
Worked out from this book	worse at 1.25x; better at 0.80x (the outcome gap luck alone can make in a pocket of typical size)
The allowance for many tests covers	each grid and measure on its own, one comparison at a time (the firm's call, 25 Sep 2026)
Revenue counts as more or less at	1.25x and 0.80x: the same lines as for losses (no pocket was big enough to work out what luck can do)
How old a loan must be to count	Every loan
Fewest loans in a pocket before it is tested	3000
Fewest loans with a loss before a loss rate is tested	10 losses
Smallest excess loss worth reporting	1% of the book's total losses
What a pocket is judged against	The rest of its band
How much worse than its comparison a pocket must be	1.25 times
How much better than its comparison a pocket must be	0.8 times
How far revenue must move before it counts as more or less	What luck alone can move it
How sure a difference must be before it counts	95% sure
How often a real gap should be caught	80% of the time
Allowing for testing many pockets at once	Hold down the share of lucky finds (Benjamini-Hochberg)

Materiality line: RANR per booked dollar	8,288 RANR_AMT dollars
Worked out from this book	worse at 1.25x; better at 0.80x (the outcome gap luck alone can make in a pocket of typical size)
The allowance for many tests covers	each grid and measure on its own, one comparison at a time (the firm's call, 25 Sep 2026)
Revenue counts as more or less at	1.25x and 0.80x: the same lines as for losses (no pocket was big enough to work out what luck can do)
How old a loan must be to count	Every loan
Fewest loans in a pocket before it is tested	3000
Fewest loans with a loss before a loss rate is tested	10 losses

How this was walked

On a Linux test machine:

- **The window:** Origination Cube.pyw's window on a virtual display under Python 3.12, its buttons pressed by driver/drive.py (through driver/win.sh), each state photographed.
- **Excel's part:** done cell by cell with openpyxl (driver/edit.py), standing in for picking from the dropdowns and typing.
- **The workbook:** each state printed through LibreOffice (driver/render.sh) and photographed page by page. driver/shots.py crops and rings every picture.
- **Wrong turns and revenue options:** each in a folder of its own, from a copy of the answered workbook (driver/wrong.sh). They share one scratch memory file with the main route, as a team sharing a memory folder would.
- **The build:** frozen at commit 5ef61df with git archive.
- **Reading tabs back:** driver/boxcheck.py, driver/pastline.py (dots past a line but boxed "the same"), driver/revenue_options.py (step 13's table), driver/threeway.py, driver/luckline.py (what the suggested numbers are made of), driver/noplant.py (the split on a book with no debt effect).
- **The next extracts:** synth.make_rows with seeds 11 (Q4, 6,000 loans) and 23 (Q1, 5,000 loans).

The order walked: steps 1 to 17, the revenue options, 18 to 21 on the same workbook, step 24 and the band-width wrong turns on the Q4 extract, then 22 and 23, then the other wrong turns. That order is why step 22 picked up edges from the Q4 wrong turns.